PP

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PLATING POSTERS INC · METAL FINISHING REFERENCE SERIES

CHROMIC ANODIZING

TYPE I

ANODIZE CLASS **TYPE I**

TYPE I	TYPE II	TYPE III
CHROMIC · THIN	SULFURIC · ~5–25 MM	HARDCOAT · THICK

CHROMIC · THIN · AEROSPACE · CR VI

THE COMPLETE TRAINING MANUAL

*A full training course for the brand-new operator — from "wait, the part is the positive electrode?" to a signed certificate of completion. Thin, gray, corrosion-protecting chromic-acid anodize on aluminum — the aerospace finish that protects fatigue-critical parts and primes them for paint. We don't deposit metal here — we **grow** oxide out of the part. And the bath is **hexavalent chromium, a confirmed human carcinogen** — so this manual teaches the chemistry and the most dangerous safety discipline on any anodize line. Assumes you know nothing. Teaches you everything.*

01 INSP/RACK ▸ 02 CLEAN ▸ 03 ETCH ▸ 04 RINSE ▸ 05 DESMUT ▸ 06 RINSE

07 ANODIZE ▸ 08 RINSE ▸ 09 DYE ▸ 10 SEAL ▸ 11 RINSE/DRY ▸ 12 INSPECT

OPERATOR TRAINING & CERTIFICATION

EDITION 1.0 · 2026 · FOR-DUMMIES STYLE

FIND YOUR WAY

TABLE OF CONTENTS

READ FRONT-TO-BACK THE FIRST TIME · KEEP ON THE SHELF FOREVER AFTER

PART ONE — FRONT MATTER & FUNDAMENTALS

PART TWO — STATION BY STATION

PART THREE — ACROSS THE LINE

PART FOUR — CERTIFICATION

★ **REMEMBER**

This manual is a training and reference tool — **not** a substitute for your facility’s written procedures, your supplier’s TDS/SDS, your customer’s spec (MIL-PRF-8625 Type I, AMS, ASTM), your shop’s OSHA Cr(VI) compliance program, or your supervisor’s instructions. When this manual and your shop’s official procedure disagree, **follow your shop’s official procedure and ask your supervisor.**

PART ONE — FRONT MATTER & FUNDAMENTALS

START HERE

WELCOME ABOARD

IF YOU DON'T KNOW AN ANODE FROM AN APRON, YOU'RE IN EXACTLY THE RIGHT PLACE — AND HERE, THE BATH ITSELF IS A CARCINOGEN, SO THE STAKES ARE YOUR HEALTH, NOT JUST QUALITY

Welcome. If you are holding this manual, you are about to learn how to run — or at least understand — a **chromic acid anodizing line** (Type I). Maybe you started this week. Maybe you've been racking parts for a year and finally want to know *why* you do the things you do. Either way, you're in the right place, and we're glad you're here.

Here is the first big thing to wrap your head around, and it's going to feel backwards if you've ever seen electroplating: **in anodizing, we are not coating your part with a different metal. We are growing a tough, ceramic oxide layer *out of the aluminum itself*.** And to do that, your part is hooked up as the **anode** — the *positive* electrode — which is the exact opposite of plating. If that sentence made your eyebrows go up, good. That's the single biggest *process* idea in this whole book.

But there is a second idea that matters even more than the first, and we will repeat it more than any other line in this series: **the chromic anodize bath is hexavalent chromium — Cr(VI) — a confirmed human carcinogen.** It is the most hazardous bath of the three anodize types. This is not said to scare you off. A trained, careful operator who respects this chemistry can work it safely for an entire career — an untrained or careless one cannot. This manual teaches that respect, station by station.

The third thing we'll tell you up front: **this manual assumes you know nothing about anodizing, chemistry, or electricity.** That's not an insult — it's a promise. Every term gets defined the first time we use it; every step explains not just *what* to do but *why* it matters. It's written in the spirit of the *For Dummies* books — friendly, plain-spoken, and patient. We would rather over-explain than leave you guessing in front of a tank full of chromic acid.

> ★ **REMEMBER**
> You do not need to memorize this manual. You need to *understand* it. Understanding sticks. Memorizing fades by Friday — and on a Cr(VI) line, the understanding is what keeps you healthy.

• HOW TO USE THIS MANUAL

The manual follows the **line itself**, station by station, in the order a part actually travels: inspect and rack it, clean it, lightly etch it, rinse it, desmut it, rinse it, **anodize it** (in chromic acid, part = anode, ramped voltage), rinse it, (almost never dye it), seal it, rinse/dry it, and inspect it. Reading it in order matters, because **the order of an anodizing line is not random** — and neither is the order of this book.

> 💡 **TIP**
> Read the Fundamentals (Part One) all the way through *before* you read any individual station chapter — **especially Chapter 3 on the Cr(VI) hazard.** The station chapters assume you already understand the big picture: the part is the anode, oxide grows out of it, and the bath is a carcinogen whose worst hazard is an invisible mist.

THE ICONS — YOUR FRIENDLY MARGIN HELPERS

> 💡 **TIP**
> A shortcut or trick of the trade — the kind of thing a 20-year veteran would lean over and tell you.

> ★ **REMEMBER**
> A core idea worth locking in. The load-bearing ideas.

> ⚠ **HEADS-UP**
> A safety warning. Read every single one. On a chromic line these protect your lungs, skin, eyes, nose, and long-term health. We use them *very* liberally — more than the sulfuric manuals — because Cr(VI) earns it.

> ⚙ **TECHNICAL STUFF**
> Deeper chemistry for the curious. Optional.

> ☀ **FUN FACT**
> A bit of trivia to make the science stick (and to give you something to say at lunch).

TWO MORE THINGS YOU'LL SEE AT EVERY STATION

Each station chapter ends with a **Teach-Back** checklist (often paired with a red "Top Mistakes" card) — the things you must say out loud, unprompted, before you're cleared — and a bordered **Sign-Off** box your *trainer initials* once you've demonstrated the skill safely. Treat the Teach-Back as your study guide and the Sign-Off as the gate you must pass.

WHAT YOU'LL BE ABLE TO DO

LEARNING OBJECTIVES

MASTER THIS AND EVERY STATION CHAPTER CLICKS INTO PLACE

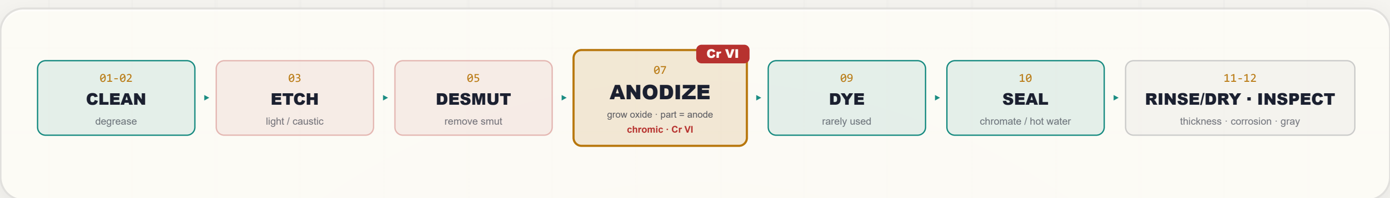
By the time you finish this manual, you should be able to confidently:

1. **Explain what anodizing is** in plain language — and how it is fundamentally *different from electroplating*: the part is the **anode**, and we **grow an aluminum-oxide layer out of the aluminum itself** instead of depositing a different metal onto it.
2. **State the central safety fact of this line:** the bath is **hexavalent chromium (Cr VI) chromic acid, a confirmed human carcinogen**, whose worst hazard is an *invisible mist* — and name the controls that keep you safe (mist suppression, ventilation/scrubber, respirator, hygiene, no eating/smoking).
3. **Describe what Type I chromic anodize does** and why a shop chooses it — a **thin, soft, gray/opaque** corrosion-protective oxide that **barely changes dimensions, does not significantly reduce the aluminum's fatigue strength, leaves only mild residue in crevices, and primes beautifully for paint and adhesive bonding.**
4. **Explain why Type I is the *aerospace* anodize** — fatigue-critical structural parts, assemblies, lap joints, faying surfaces, and paint/primer bases.
5. **Explain the thin porous oxide** — why anodic oxide has pores, why Type I is usually left **natural or used as a paint base** rather than dyed, and how it is **sealed** (dilute chromate or hot water).
6. **Explain dimensional growth** — the coating grows roughly half in/half out, but because Type I is *thin* (~0.5–7.5 μm), the change is tiny — a key reason it's specified on tight-tolerance parts.
7. **Place Type I in the family** — Type I (chromic), Type IB (low-voltage chromic), Type IC (non-chromic substitute, e.g., BSAA), Type II (sulfuric), Type III (hardcoat) — and understand the **industry phase-down** of hex-chrome toward **BSAA** and **thin-film sulfuric (TSA)**.
8. **Explain why aluminum alloy matters** — including the copper-rich aerospace alloys (2024, 7075) chromic is often run on.
9. **Name the stations of the line in order** and explain *why* the order can't be rearranged.
10. **Recognize the main hazards** — Cr(VI) chromic acid and mist (the headline), hot caustic etch, hydrogen gas at the cathode, DC electrical/arcing, hot seal — and know the correct PPE, emergency response, and **reduce-then-precipitate** Cr(VI) waste handling.

> ★ **REMEMBER**
> You are not expected to hit all of these on day one. Anodizing is a craft. But the *safety* objectives (#2) are not optional and not gradual — you must have those before you work the line at all.

• THE LINE AT A GLANCE

Don't memorize yet — feel the rhythm: **Inspect/Rack → Clean → Etch → Rinse → Desmut → Rinse → ANODIZE (chromic) → Rinse → (Dye, rarely) → Seal → Rinse/Dry → Inspect.**



> ★ **REMEMBER**
> Notice the pattern — **a rinse follows almost every working tank.** On a chromic line, rinses do double duty: they keep one chemistry from poisoning the next *and* they capture dragged-out **carcinogenic chromium** so it doesn't spread around the shop or reach a drain. They aren't filler; they're some of the most important steps you'll perform.

> ☀ **FUN FACT**
> The aluminum oxide you grow in this tank is chemically the same family as **sapphire and ruby** (crystalline aluminum oxide, Al_2O_3). You're not bolting on a coating — you're turning the skin of the part into a ceramic that's part of the metal.

CHAPTER 1

WHAT IS ANODIZING?

THE TRUE BEGINNER'S PRIMER — SO FAR BACK WE DON'T EVEN ASSUME YOU KNOW WHAT ANODIZING IS

THE ONE-SENTENCE VERSION

Anodizing is using electricity to grow a hard, protective oxide layer out of the surface of an aluminum part — the part itself becomes part of the coating. That's the whole idea. It's genuinely different from plating, and that difference trips up almost everyone at first.

FIRST, WHAT PLATING DOES (SO YOU CAN SEE THE DIFFERENCE)

In electroplating — zinc, nickel, chrome — you take your part, hook it up as the **cathode** (the *negative* electrode), and you *deposit a different metal* onto it from a bath. The coating is a *separate metal sitting on top of* your part.

Anodizing flips almost all of that:

- Your part is the **anode** — the **positive** electrode. (That's literally where the word "anodize" comes from.)
- We don't add a different metal. We take the **aluminum that's already there** and convert its surface into **aluminum oxide** by feeding electricity through an acid bath.
- The coating isn't sitting *on* the part — it **is** the part's surface, chemically grown and locked in. It can't peel like paint or flake like a bad plate, because it's integral to the metal.

> ★ **REMEMBER — THE SINGLE BIGGEST *PROCESS* IDEA IN THIS BOOK**
> In plating, the part is the **cathode** and you **add** a metal. In anodizing, the part is the **anode** and you **grow oxide out of the part itself.** Everything else follows from it.

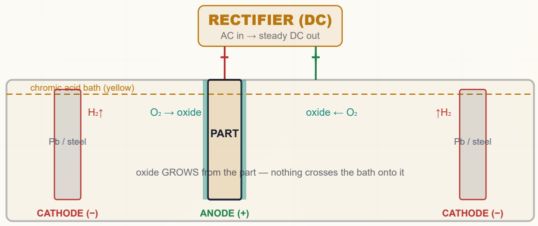
HOW IT PHYSICALLY WORKS: THE KITCHEN-SINK PICTURE

Picture a tank of liquid called the **electrolyte** (or "the bath"). For Type I, that liquid is **chromic acid** — chromium trioxide (CrO_3) dissolved in water, a yellow-to-orange solution. You hook two things to a power supply called a **rectifier** (it converts ordinary AC wall power into the steady one-direction "DC" current anodizing needs):

- **The anode** — this is your **aluminum part**, connected to the **positive** terminal. *"Anode = the part, the thing growing oxide."*
- **The cathode** — bars or plates (commonly **steel or lead**) hanging in the same tank, connected to the **negative** terminal. (In chromic anodizing a steel tank can itself serve as the cathode.) The cathode just completes the circuit and is where **hydrogen gas** bubbles off; it does **not** feed metal into the part.

When you turn on the rectifier:

1. Current flows. At the surface of your aluminum part (the anode), water in the bath is split and **oxygen** is delivered right at the metal surface.
2. That oxygen immediately combines with the surface aluminum to form **aluminum oxide (Al_2O_3)** — a hard, ceramic layer.
3. The chromic acid is mildly aggressive, so as the oxide forms, the acid is *also* gently dissolving it — and this tug-of-war creates the **porous structure** (Chapter 4).
4. Over at the cathode, **hydrogen gas** bubbles off (flammable — a safety item we'll revisit).



Compare to a plating cell: the labels are reversed — here the **part is the ANODE (+)** and the counter-electrodes are the cathodes.

TECHNICAL STUFF — THE TWO REACTIONS

At the part (anode), aluminum is oxidized: simplified, $2Al + 3H_2O \rightarrow Al_2O_3 + 6H^+ + 6e^-$. At the cathode, those electrons reduce hydrogen ions: $6H^+ + 6e^- \rightarrow 3H_2\uparrow$. So **oxygen builds the coating at your part; hydrogen gas leaves at the counter-electrode.** No aluminum travels through the bath onto something else — it stays put and turns to oxide in place.

TECHNICAL STUFF — WHY TYPE I RUNS *WARM*, NOT COLD

Unlike a sulfuric Type II tank (held cold, ~20 °C), chromic anodize runs **warm — ~32–40 °C (90–105 °F).** Chromic acid is a *gentler* electrolyte than sulfuric, so it dissolves the growing oxide more slowly; the warmth is what lets the oxide form at a workable rate and gives chromic its characteristic thin, dense coating. The trade-off: a warm chromic-acid tank throws **more mist** — which on a Cr(VI) bath is the central hazard (Chapter 3).

FUN FACT

Anodizing is one of the few finishing processes where the coating is *grown from* the part rather than *added to* it. That's why anodized aluminum can't "chip off" to bare metal the way paint or a bad plate can — there's no glue line. The oxide and the metal are continuous.

CHAPTER 2

WHAT IS “TYPE I”?

THE THIN, GRAY AEROSPACE ANODIZE — IT BARELY CHANGES THE PART, IT’S KIND TO FATIGUE LIFE, AND IT’S THE BEST PAINT BASE IN THE FAMILY

Type I anodizing is the **chromic-acid anodize**: a thin, dense aluminum-oxide layer, typically **~0.5–7.5 µm (most often ~2–5 µm)**, grown in a warm chromic-acid bath with a **ramped DC voltage**. The coating is **thin, soft, and gray/opaque** — it is **not** the general decorative coat (that’s Type II sulfuric) and it is **not** for wear (that’s Type III hardcoat). Its niche is **corrosion protection plus an excellent paint/primer and adhesive-bond base, especially on aerospace parts.** The “Type I” name comes from the U.S. spec **MIL-A-8625 / MIL-PRF-8625**, which sorts anodize into Type I (chromic), Type II (sulfuric), and Type III (hardcoat).

• WHY AEROSPACE SPECIFIES IT

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Doesn’t hurt fatigue strength	Chromic anodize **does not significantly reduce the fatigue strength** of the aluminum the way sulfuric can. Specified on **fatigue-critical structural parts** — spars, ribs, fittings, landing-gear components.
Safe in crevices & assemblies	Any chromic-acid **residue trapped** in crevices, **lap joints, spot-welds, faying surfaces, or blind holes is far less aggressive/corrosive** than trapped sulfuric. Preferred for built-up assemblies and faying (mating) surfaces.
Minimal dimensional change	The coating is *thin*, so the part barely grows — ideal for **tight-tolerance** parts where Type II/III growth would be a problem.
Excellent paint/primer base	The thin porous oxide is one of the best **adhesion bases for primers, paints, and structural adhesives** in the whole anodize family.
Corrosion protection	Even thin and sealed in chromate, it protects the aluminum and carries corrosion-inhibiting chromate into the coating.

WHERE YOU’LL SEE IT

Aircraft structure and fittings, fatigue-critical machined parts, riveted/spot-welded assemblies, faying surfaces in airframes, and parts that will be **primed and painted** or **adhesive-bonded**. When a print calls “MIL-A-8625 Type I” (or AMS equivalents), it almost always means an aerospace part.

★ **REMEMBER**

Type I’s superpowers are **(1) it protects without hurting fatigue life, (2) it’s gentle in crevices/assemblies, (3) it barely changes dimensions, and (4) it’s a superb paint/bond base.** What it is *not*: thick, hard, wear-resistant, or brightly dyeable. If a customer wants color or wear, they want Type II or Type III — not Type I.

💡 **TIP**

When a print says just “anodize,” it usually means **Type II sulfuric.** “Type I,” “chromic,” “MIL-A-8625 Type I,” or an aerospace AMS callout means **this** process — different chemistry, much thinner coating, and a **carcinogenic bath.** Always confirm the type before you run.

⚠ **HEADS-UP**

The thing that makes Type I special on an aircraft (gentle, thin, fatigue-friendly) comes at a price you carry personally: the electrolyte is **hexavalent chromium, a confirmed carcinogen.** That is why industry is phasing it down (Chapter 6) — and why the very next chapter is the most important one in this book.

☀ **FUN FACT**

The original chromic anodize process — the **Bengough–Stuart process**, developed in Britain in the 1920s — was one of the *first* commercial anodizing methods ever used, and a version of its voltage cycle is still written into aerospace specs a century later.

CHAPTER 3 · THE MOST IMPORTANT CHAPTER

WHY HEXAVALENT?

READ THIS CHAPTER TWICE. ON THIS LINE YOUR HEALTH, NOT JUST YOUR PARTS, DEPENDS ON IT

We put this chapter near the front on purpose. On the sulfuric manuals, the porous-oxide chapter can come first. **Not here.** The defining feature of chromic anodize is that it runs on **hexavalent chromium, Cr(VI)** — and Cr(VI) is a **confirmed human carcinogen.** You need to understand this *before* you read another word about how to anodize.

WHY THE BATH IS HEXAVALENT

The Type I bath is **chromic acid** — chromium trioxide (CrO_3) dissolved in water. When CrO_3 dissolves, the chromium is in its **+6 ("hexavalent") oxidation state.** That hexavalent chromium is what makes the gentle, thin, fatigue-friendly chromic anodize possible at all. The same property that makes the chemistry desirable is what makes it dangerous.

TECHNICAL STUFF

"Hexavalent" (Cr^{6+}, Cr VI) and "trivalent" (Cr^{3+}, Cr III) refer to the chromium atom's oxidation state. Cr(VI) is a strong oxidizer and is highly toxic and carcinogenic. Cr(III) — the form chromium ends up in *after* it's reduced — is far less hazardous and is actually a trace nutrient. **The whole goal of waste treatment on this line (Part Three) is to convert the dangerous Cr^{6+} into the safe Cr^{3+} and remove it.**

HOW CR(VI) HURTS PEOPLE

This is not abstract. Cr(VI) exposure on an anodize line causes real, documented harm:

- **Lung cancer** — the most serious. Long-term inhalation of chromic-acid mist is linked to lung and sinonasal cancer. This is why the bath is regulated as a carcinogen.
- **Nasal septum ulceration and perforation** — chromic-acid mist literally eats a hole through the cartilage wall between the nostrils. Once a common, visible mark of long-time chrome workers — and **preventable** with proper controls.
- **"Chrome holes" / chrome ulcers** — deep, slow-healing skin ulcers (often on hands, knuckles, forearms) where chromic acid contacts even a small break in the skin.
- **Skin and respiratory irritation, dermatitis, and allergic sensitization.**
- **Eye damage** — chromic acid is corrosive to the eyes.

HEADS-UP — THE SINGLE MOST DANGEROUS CHEMISTRY ON ANY ANODIZE LINE

The hazard is mostly **invisible.** The chromic-acid **mist/aerosol** thrown off a warm, gassing tank is the main route into your lungs, and you can be over-exposed without seeing or strongly smelling anything. **Engineering controls (mist suppression + ventilation) are your primary protection — not just PPE.** If the ventilation or fume suppressant is not working, the tank is not safe to run. Stop and report it.

THE LAW: OSHA'S CR(VI) STANDARD

Cr(VI) has its own dedicated OSHA standard — **29 CFR 1910.1026.** You don't need to memorize the legal text, but you must know the numbers your shop's program is built around:

TERM	VALUE	PLAIN MEANING
PEL (Permissible Exposure Limit)	5 µg/m³	The legal ceiling for Cr(VI) in the air you breathe, averaged over an 8-hr shift
Action Level	2.5 µg/m³	The "pay attention" trigger (8-hr avg) — at or above this, the shop must do air monitoring, training, and medical surveillance

A microgram (µg) is a *millionth* of a gram. A limit of 5 µg per cubic meter of air is *extraordinarily* small — which tells you how potent this hazard is and why mist control is non-negotiable.

HEADS-UP — YOU CAN'T SMELL YOUR WAY TO SAFETY

Because the PEL is so low, you **cannot judge your exposure by smell or eye irritation** — those happen at far higher levels. That's why OSHA requires *air monitoring*, *medical surveillance*, regulated areas, a written exposure-control plan, and specific hygiene facilities. Your shop's Cr(VI) program is a legal requirement, not a suggestion.

• THE FIVE PILLARS OF CR(VI) CONTROL

You'll see these at every station on the line:

1. **Mist suppression in the tank** — a **fume suppressant / foam blanket** (a surfactant additive that forms a foam layer and lowers surface tension so bursting gas bubbles throw far less mist), and/or floating plastic balls. Your first line of defense, right at the source.
2. **Tank ventilation** — **local exhaust** (push-pull or lateral-exhaust hoods) that pulls mist off the tank surface, plus a **fume scrubber** that washes Cr(VI) out of the exhausted air before it leaves the building.
3. **Respiratory protection** — the right respirator for the task, per your shop's program, used *in addition to* (never instead of) ventilation.
4. **Hygiene and no hand-to-mouth** — **no eating, drinking, smoking, vaping, or cosmetics** anywhere near the line; designated wash-up; separate work clothing; decontamination before breaks and before leaving.
5. **Spill, waste, and skin protection** — full chemical PPE, immediate flushing of any skin/eye contact, controlled spill response, and proper **Cr(VI) waste treatment** (reduce $Cr^{6+} \rightarrow Cr^{3+}$, then precipitate — Part Three).

> ★ **REMEMBER**
>
> Engineering controls first (suppress the mist, ventilate the tank), respirator second, hygiene always. PPE is the *last* layer of defense, not the first. A respirator does not make a tank with broken ventilation safe.

> ⚠ **HEADS-UP — HYGIENE IS PART OF THE JOB, NOT OPTIONAL**
>
> Hand-to-mouth contact is a real route for getting Cr(VI) into your body. **Never** eat, drink, smoke, vape, or touch your face on the line. Wash thoroughly before every break and before going home. Keep work clothing and street clothing separate so you don't carry chromium home to your family.

> ☀ **FUN FACT**
>
> The yellow-orange color of the chromic-acid bath comes from the hexavalent chromium itself. When waste treatment reduces it to trivalent, the color shifts toward green — operators can sometimes *see* the reduction reaction happening as the color changes. (Color is a clue, never a substitute for proper testing.)

> ★ **REMEMBER — IF YOU TAKE ONLY ONE THING FROM THIS CHAPTER**
>
> The chromic-acid bath is a **confirmed human carcinogen** whose worst hazard is an *invisible mist.* Suppress the mist, ventilate the tank, wear your respirator, keep your hands away from your face, and never let Cr(VI) reach a drain. Do those things, every shift, and you can work this line safely for a career.

CHAPTER 4

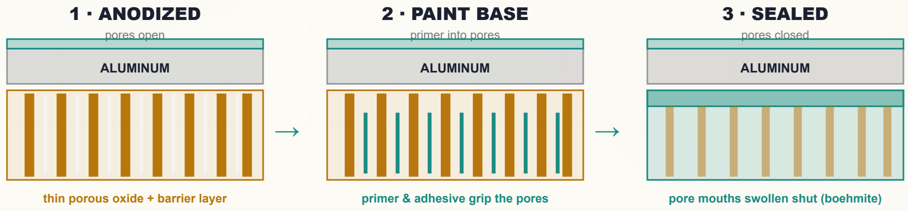
THE THIN OXIDE

SAME POROUS-OXIDE IDEA AS EVERY ANODIZE — BUT THIN, GRAY, AND BUILT TO BE PRIMED AND PAINTED, NOT COLORED

Like all anodizing, chromic anodize grows a **porous oxide:** billions of tiny oxide columns, each with a microscopic pore down the center, sitting on a thin dense **barrier layer** against the metal. Those open pores are what make *any* anodize able to absorb dye, seal, or paint.

But Type I is different from Type II in three ways that change how you treat it:

- **It's thin** (~0.5–7.5 µm vs. ~5–25 µm for Type II). Fewer, finer pores; less depth to hold color.
- **It's gray/opaque, not clear.** Even on good alloys the chromic coating is a translucent gray, so it can't carry bright, true colors the way a clear sulfuric coat can.
- **Its job is a paint base, not a color.** Type I parts are usually left **natural** or sent on to **primer/paint/adhesive** — the porous oxide grips coatings extremely well.



SO WHERE DOES DYE FIT?

For Type I, **dyeing is the exception, not the rule.** Because the coating is thin and gray, dyed Type I looks muted and is rarely specified. (When a colored chromic part is wanted, it's usually a dark shade like black, and even then many shops use Type II instead.) On most chromic lines, the part goes **anodize** → **rinse** → **seal**, skipping dye entirely. We keep a dye station in the line (Station 09) so the sequence parallels the rest of the family — but on a Type I line it usually sits idle.

SEALING A TYPE I COATING

Even unpainted Type I is usually **sealed** to close the pores and add corrosion protection. The two common routes: a **dilute chromate (dichromate) seal** (a hot dilute sodium-dichromate solution that closes pores *and* deposits corrosion-inhibiting chromate — but is **itself Cr(VI)**), and a **hot DI water seal** (near-boiling water hydrates the pore-wall oxide to **boehmite**, swelling the pores shut; metal-free).

★ **REMEMBER — THE HEART OF A TYPE I LINE**

Anodize → rinse → (rarely dye) → Seal. Anodize grows the thin gray porous oxide; the part is usually left natural or primed/painted; sealing (chromate or hot water) closes the pores and locks in corrosion resistance. Dye is the exception, not the rule.

⚠ **HEADS-UP — THE DICHROMATE SEAL IS *ALSO* HEXAVALENT CHROMIUM**

Don't relax at the seal tank just because the anodize is done. A sodium-dichromate seal is **Cr(VI)** — same mist, skin, and waste hazards as the anodize bath. Hot-water seal avoids that, which is one reason many shops prefer it.

☀ **FUN FACT**

An unsealed, freshly anodized part is so absorbent that you can write on it with a permanent marker and the "ink" soaks in. The same thirst is what makes the oxide drink up primer — and why a sealed part, with its pores closed, no longer takes paint as readily (paint goes on *before* a full seal, or onto a deliberately lightly-sealed coat, per the spec).

CHAPTER 5

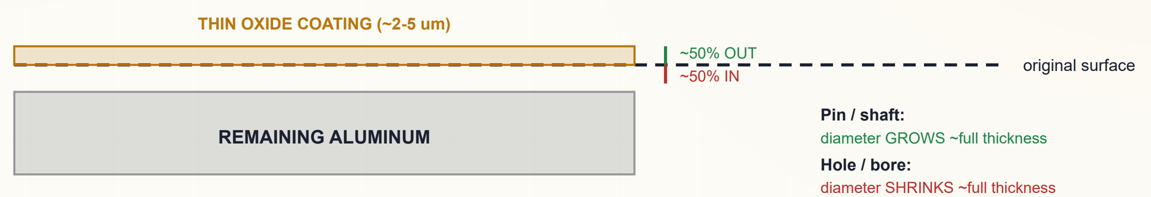
DIMENSIONAL GROWTH

THE COATING STILL GROWS HALF IN, HALF OUT — BUT IT'S SO THIN THAT ON TYPE I THE CHANGE IS TINY, AND THAT'S EXACTLY WHY IT'S SPECIFIED

Because we're converting aluminum into aluminum oxide, and oxide takes up **more room** than the metal it came from, the coating grows in **two directions at once**:

- About **half** the coating thickness grows *into* the original surface (consuming aluminum).
- About **half** grows *out* from the original surface (the part gets bigger).

The physics is identical to Type II — but the *numbers* are small because Type I is thin. For a typical **~2–5 µm** chromic coat, each surface moves out only about **~1–2.5 µm**, and a diameter changes by only about **the full coating thickness (~2–5 µm).** That's often within the tolerance band of a precision part — which is precisely why designers choose chromic for **tight-tolerance features** where Type II (~5–25 µm) or Type III (~25–100+ µm) growth would push parts out of spec.



Type I growth is tiny — often within tolerance — but it's still ~half in / half out, not zero.

★ REMEMBER — THE 50% RULE, WITH TINY NUMBERS

Anodize grows **roughly half in, half out** regardless of type. Each surface moves out about **half** the coating thickness; a diameter changes by about the **full** thickness. On Type I the coating is so thin that the change is often negligible — but on truly critical fits and threads you still **account for it or mask the area.**

💡 TIP

Even though Type I growth is small, don't tell a customer "zero." If a feature is critical (a bearing fit, a thread, a gauge surface), confirm whether it should be **masked**, machined to allow for growth, or measured after anodizing. "Small" is not the same as "none."

⚠ HEADS-UP — MASKING AND CONDUCTIVITY

Areas that must *not* grow or must stay electrically conductive (threaded holes, electrical-ground points, bearing fits, faying surfaces that get a *different* treatment) are **masked off** so no oxide forms there. Anodic oxide is **electrically insulating**, so any surface that must conduct after anodizing has to be masked or it won't work.

⚙ TECHNICAL STUFF — THE PILLING–BEDWORTH IDEA

Aluminum oxide occupies more volume than the aluminum it formed from (Pilling–Bedworth ratio > 1). That volume expansion is *why* the coating grows outward at all, and the same expansion lets hot-water sealing pinch the pores shut. One fact — oxide is bulkier than metal — explains both dimensional growth *and* sealing.

☀ FUN FACT

Type III hardcoat grows so much that aerospace drawings call out separate pre- and post-anodize dimensions. Type I is the opposite extreme in the same family — so thin that the same drawing might not need a growth allowance at all. Same physics; opposite ends of the ruler.

CHAPTER 6

THE TYPE FAMILY

SAME IDEA, SEVERAL FLAVORS — AND AN INDUSTRY ACTIVELY MOVING AWAY FROM THE CHROMIC ONE

All anodize types grow an integral aluminum-oxide coating with the part as the anode. They differ in **acid, thickness, voltage, and purpose.**

	TYPE I — CHROMIC *(THIS MANUAL)*	TYPE II — SULFURIC	TYPE III — HARDCOAT
Electrolyte	**Chromic acid (CrO_3) ~30–100 g/L**	Sulfuric acid (~165–200 g/L)	Sulfuric (concentrated, colder)
Typical thickness	~0.5–7.5 µm (often 2–5)	~5–25 µm	~25–100+ µm
Temperature	**Warm ~32–40 °C (90–105 °F)**	~20–21 °C	Cold ~0–10 °C
Voltage / current	**Ramped/stepped ~22→40 V; low CD**	~12–18 V, ~12–18 ASF	Higher V, higher CD
Color / dyeing	**Gray/opaque; rarely dyed**	Excellent dyeing	Darkens; bronze/black
Main use	**Aerospace — fatigue-safe, crevice-safe, tight-tolerance, paint base**	Decorative + protective workhorse	Maximum hardness & wear
Headline hazard	**Cr VI carcinogen (worst of the three)**	Sulfuric acid burns	Sulfuric acid, cold bath

THE SUB-CLASSES YOU'LL MEET ON PRINTS

- **Type I** — conventional chromic anodize, the ramped cycle holding around **~40 V** (Bengough–Stuart lineage).
- **Type IB** — **low-voltage chromic**, a cycle that holds around **~22 V**; a defined alternative within MIL-PRF-8625 for certain alloys/needs.
- **Type IC** — a **non-chromic substitute for Type I**, developed specifically to *replace* hex-chrome while giving Type-I-like thin coatings and properties. The most common Type IC process is **BSAA (Boric-Sulfuric Acid Anodize).**

★ **REMEMBER**

Type I = chromic, thin, aerospace/fatigue/paint-base, Cr VI. Type IB = low-voltage chromic (~22 V). Type IC = non-chromic substitute (BSAA). Type II = sulfuric decorative/protective. Type III = sulfuric hardcoat, thick/hard. This manual is Type I (and IB) — the *chromic* members of the family.

THE INDUSTRY TREND: PHASING DOWN HEX-CHROME

Because Cr(VI) is a carcinogen and is tightly regulated (OSHA 1910.1026; REACH authorization in the EU), the industry is **actively reducing chromic anodize.** The two common replacements: **BSAA (Boric-Sulfuric Acid Anodize, Type IC)** — a thin sulfuric-based coating with a little boric acid, run at low voltage, giving **Type-I-like thinness, low dimensional change, and a good paint base — with no hexavalent chromium** (the leading drop-in replacement, widely qualified on aerospace primes); and **Thin-Film Sulfuric Anodize (TSA)** — a thinned-down, lower-voltage sulfuric process used as another chromic-free alternative for corrosion protection and paint base.

💡 **TIP**

If your shop is moving toward **BSAA or TSA**, the *handling* of parts (rack, clean, etch, rinse, seal, inspect) is very similar to what you'll learn here — but the **main bath is no longer a carcinogen.** That's the whole point of the switch. We build the **BSAA** manual next in this series.

⚙ **TECHNICAL STUFF — WHY CHROMIC WAS WORTH THE HAZARD FOR SO LONG**

Chromic anodize is thin, its acid is *gentle* on the base metal (so it removes little aluminum and is kind to **fatigue life**), and any trapped residue is far less corrosive than trapped sulfuric. Those are real engineering advantages — which is why aerospace specs leaned on it for decades despite the Cr(VI) hazard, and why replacements like **BSAA** had to be carefully qualified to match those properties before chromic could be retired.

CHAPTER 7

ALUMINUM ALLOYS

ANODIZING ONLY WORKS ON ALUMINUM — AND THE AEROSPACE ALLOYS CHROMIC LOVES ARE THE TRICKIEST ONES

Anodizing only works on **aluminum** and a few other “valve metals” (titanium, magnesium, tantalum, niobium — each with its own process). On this line it’s aluminum. But “aluminum” isn’t one thing — it’s dozens of **alloys**, and the alloying elements change how the part anodizes.

WROUGHT VS. CAST

- **Wrought alloys** (rolled/extruded sheet, plate, bar, extrusion — the 1xxx–7xxx series) generally anodize cleanly and predictably.
- **Cast alloys** (high-silicon) anodize **gray, mottled, or dull** and can be hard to coat evenly.

THE AEROSPACE REALITY: COPPER-RICH ALLOYS

Type I is most often run on **fatigue-critical aerospace structure**, and a lot of that structure is **2xxx (copper)** and **7xxx (zinc)** alloys:

- **Copper (2xxx, e.g., 2024):** copper-rich alloys anodize less brightly, give a duller coating, leave heavy **copper smut** after etching, and are more prone to defects. They still anodize, but they demand good cleaning, careful (light) etching, and thorough desmut. *Chromic is actually well-suited here* — its gentleness is part of why it’s chosen for these high-strength alloys.
- **Zinc (7xxx, e.g., 7075):** anodizes reasonably but varies; another common aerospace structural alloy.
- **Silicon (cast):** turns the coating gray/charcoal and mottled.

THE FRIENDLY ALLOYS

5xxx (magnesium) and **6xxx (Mg-Si, e.g., 6061/6063)** anodize very well. They’re less common on chromic/aerospace work than 2xxx/7xxx, but they coat cleanly when you see them.

ALLOY FAMILY	MAIN ELEMENT	ANODIZE BEHAVIOR (CHROMIC)
1xxx (pure Al)	none	Excellent, clean
2xxx	**Copper**	Dull, heavy smut — common aerospace alloy chromic is chosen for
5xxx	Magnesium	Very good
6xxx (6061/6063)	Mg + Si	Excellent
7xxx	**Zinc**	Good, variable — common aerospace structural alloy
Cast (high-Si)	Silicon	Gray/mottled, difficult

★ **REMEMBER**

The alloy decides the look and the prep. Chromic anodize is often run on **copper-rich 2xxx (2024)** and **zinc-rich 7xxx (7075)** aerospace alloys — exactly the alloys that leave heavy smut and coat dull. That’s normal for the application; the job is to clean, etch *lightly*, and desmut them well, not to make them look bright.

💡 **TIP**

Always confirm the **alloy and temper** against the traveler before you run — aerospace specs are alloy-specific, and some high-copper alloys have *restrictions or different cycles* under MIL-PRF-8625. When in doubt, ask; don’t assume one chromic recipe fits every alloy.

⚠ **HEADS-UP — HIGH-COPPER ALLOYS AND OVER-ETCHING**

Copper alloys lose strength and tolerance fast if over-etched, and over-etching a fatigue-critical part defeats the very reason you chose chromic. Keep the etch *light* (Station 03) on 2xxx/7xxx structure.

☀ **FUN FACT**

The same copper that makes 2024 a strong, fatigue-resistant aerospace alloy is what makes it *hard to anodize prettily* — copper doesn’t convert to clear oxide. Engineers happily trade appearance for strength here; on a chromic line, “dull and gray but sound” is a passing part.

CHAPTER 8

HOW THE LINE WORKS

EVERY TANK HAS A JOB, THE ORDER IS THE RECIPE — AND ON THIS LINE, EVERY RINSE ALSO CORRALS A CARCINOGEN

A Type I line is a sequence of tanks. Here's the whole journey and *why each step is where it is.*

1. **Inspect / Rack** — Check the part, confirm **alloy and spec** (Type I/IB, thickness, sealed/painted), and rack it for firm current-carrying contact. The part *is* the anode, so contact is everything; the contact point won't anodize.
2. **Clean / Degrease** — Remove oils and soil so oxide grows evenly. (Mild **non-etching** alkaline cleaner.)
3. **Etch (light)** — A *controlled, often light* caustic or proprietary etch removes the natural oxide skin and evens the surface — kept **light** on Type I to protect fatigue life and tolerance.
4. **Rinse** — Wash off caustic before the acid steps.
5. **Desmut / Deox** — Remove the dark **smut** (copper/silicon/iron left by etching) so the surface is uniform before anodizing.
6. **Rinse** — Clean before the main tank.
7. **ANODIZE (chromic)** — The main event. Part = anode, chromic acid, **warm**, **ramped/stepped voltage**, DC on. Grow the thin gray oxide. **Cr VI — the carcinogen tank.**
8. **Rinse** — Gently rinse chromic acid out of the fresh pores — and **capture the dragged-out chromium**, never to a drain.
9. **Dye (rarely used)** — Color into the pores only if a color is specified (uncommon for Type I).
10. **Seal** — Close the pores with **dilute chromate** (itself Cr VI) or **hot water**.
11. **Rinse / Dry** — Final clean-up and gentle dry.
12. **Inspect** — Thickness, seal quality / corrosion, appearance, dimensions.

★ **REMEMBER — WHY THE ORDER CAN'T MOVE**

You can't anodize over grease or smut (defective oxide). You can't dye before pores exist, or seal before you dye. And you can't skip the post-anodize rinse on a Cr(VI) line — that's where you both protect the seal *and* recover the carcinogen. The line *is* the recipe.

⚠ **HEADS-UP — THE LINE IS A TOUR OF HAZARDS, LED BY CR(VI)**

As you walk the line you pass **hot caustic** (etch), **acids** (desmut), the **chromic-acid Cr(VI) bath and its mist** (anodize — the headline), **flammable hydrogen** (cathode), **high-current DC** (rectifier/contacts), a **hot seal** (possibly chromate, also Cr VI), and **chromium-bearing rinse water** everywhere downstream of the anodize tank. Respect each tank for what *it* is — and treat everything downstream of Station 07 as Cr(VI)-contaminated.

💡 **TIP**

When you're learning, walk the empty line with your trainer and name each tank's *job* and *main hazard* out loud. If you can do that for all twelve — and flag which tanks and rinses carry Cr(VI) — the rest of the manual is mostly detail.

CHAPTER 9

PPE — YOUR LAST LAYER

YOUR LAST LAYER OF DEFENSE AGAINST A CARCINOGENIC, CORROSIVE, WARM ACID — WORN OVER GOOD ENGINEERING CONTROLS, NEVER INSTEAD OF THEM

A chromic anodize line works with hot caustic etch, acids, high-current DC, a hot seal, and — above all — **carcinogenic Cr(VI) chromic acid and its mist.** PPE is your *last* layer of protection (after mist suppression and ventilation), but it is essential.

★ **REMEMBER**

There is no part, no order, and no deadline worth your health. Scrap can be reworked. Your lungs cannot. On a Cr(VI) line that's not a slogan — it's literal.

THE MASTER PPE SET (WEAR AT ALL CHEMICAL STATIONS)

- **Chemical splash goggles** — sealed goggles, not safety glasses. Glasses leave gaps; chromic acid finds them.
- **Face shield** — *over* the goggles for any work at the anodize, etch, desmut, or seal tanks.
- **Chemical-resistant gloves** — elbow-length, rated for chromic acid and the cleaners (nitrile/neoprene/PVC per the SDS). **Inspect for pinholes before each use — a chrome ulcer starts at a single break.**
- **Chemical-resistant apron/suit** — full coverage, chest to below the knee.
- **Chemical-resistant boots** — closed, non-slip, splash-proof.
- **Dedicated work clothing** — kept separate from street clothes; never worn home. Keeps Cr(VI) out of your car and house.
- **Respiratory protection** — the respirator your shop's Cr(VI) program specifies. Required whenever you could be exposed to chromic-acid mist above the limits, and any time ventilation is degraded.

HAZARD ON THE LINE	PPE THAT PROTECTS YOU
Cr(VI) chromic acid + mist (anodize; chromate seal)	**Respirator per program + sealed goggles + face shield + gloves + apron + boots — over working ventilation/suppression**
Hot caustic etch — deep burns	Sealed goggles, face shield, elbow gloves, apron, boots
Acids (desmut) — burns, splash	Same acid PPE; know fluoride first-aid if used
Hot seal tank — scald/steam	Face shield, heat-resistant gloves, long sleeves
DC electrical / arcing at contacts	Dry hands, dry gloves, no jewelry, follow LOTO
Hydrogen gas at cathode	Ventilation on; no ignition sources
Slips on wet floors	Non-slip chemical-resistant boots

⚠ **HEADS-UP — RESPIRATOR IS A BACKUP, NOT A LICENSE**

A respirator is required, but it does **not** replace tank ventilation and mist suppression. If the fume suppressant has died (foam gone, mist visible) or the exhaust isn't pulling, the tank is unsafe even with a respirator on. **Stop and report it.** Engineering controls are the primary protection; the respirator is the last line.

💡 **TIP**

Put gear on *in order:* boots and apron first, then gloves, then goggles, then face shield and respirator last. Take it off in reverse, and **rinse your gloves *before* removing them.** This keeps contaminated gloves away from your face — critical with a carcinogen.

⚠ **HEADS-UP**

Never eat, drink, smoke, vape, or apply cosmetics on the line. Hand-to-mouth contact is a major route for ingesting Cr(VI). Wash thoroughly before breaks and before leaving. Keep food and drink completely out of the work area, and work clothes separate from street clothes.

CHAPTER 10

EMERGENCY & FIRST-RESPONSE

KNOW THESE COLD BEFORE YOU TOUCH A TANK — ON A CR(VI) LINE, SECONDS AND REPORTING BOTH MATTER

Know these *before* you need them. **Locate, before your first shift:** the nearest **emergency eyewash** (within ~10 seconds of every chemical station), the **safety shower**, the **fire extinguisher** and alarm, the **rectifier emergency stop / disconnect**, the **spill kit**, the **SDS binder**, and your shop's **Cr(VI) regulated-area** boundaries and decon station.

⚠ **HEADS-UP**

Eyewash and shower stations must always be unblocked. Never stack parts, totes, or anode boxes in front of them. The day you need one is the day you can't move boxes out of the way.

SITUATION	WHAT TO DO — IMMEDIATELY
Chromic acid / chemical on skin	Safety shower; flush **at least 15 minutes.** Remove contaminated clothing while flushing. Even small chromic-acid contact on broken skin needs medical attention (**chrome-ulcer** risk). Bring the SDS.
Chromic acid / chemical in eyes	Eyewash; hold eyelids open and flush **at least 15 minutes**, rolling the eyes. Don't rub. Get medical help immediately — chromic acid is corrosive to eyes. Bring the SDS.
Inhaled chromic-acid mist, feel ill	Move to fresh air immediately. Report it — don't "tough it out." Seek medical attention for anything beyond brief mild irritation. Report as a possible **Cr(VI) exposure** so monitoring/medical follow-up happens.
Cr(VI) swallowed	**Do not** induce vomiting unless the SDS directs it. Follow SDS first aid; call poison control / medical help immediately.
Nosebleed / sore nasal passages	Report it — recurrent nosebleeds or nasal soreness can be an early sign of Cr(VI) over-exposure. Don't ignore it; it triggers a ventilation/exposure check.
Hot caustic etch or hot seal burn	Cool with water; don't pop blisters; get medical attention for anything beyond minor. Caustic burns can be deep even when they don't hurt at first.
Electrical shock / arc at rectifier	**Do not touch the person** if they may still be in contact with current. Cut power at the e-stop/disconnect first, then call for help and render aid.
Chemical spill	Alert others; keep people away; clean only if trained and within your shop's "small spill" limits. **Never let chromic-acid spills or rinse water reach a floor drain** — that's a reportable environmental release.
Fluoride exposure (if a fluoride desmut is used)	Fluoride burns are deep, delayed, and systemically toxic — flush, then apply the **specific first-aid** in the SDS (often **calcium gluconate gel**) and get medical help immediately. Ordinary flushing alone may not be enough.

⚠ **HEADS-UP — LOCKOUT/TAGOUT (LOTO)**

Mandatory for *all* rectifier and electrical maintenance. Physically lock the power off and tag it so no one can re-energize while someone works. "I'll just be a second" is how people get electrocuted.

⚠ **HEADS-UP — THE GOLDEN RULE FOR ACID MAKE-UP**

Always add ACID/chemical to WATER, never water to concentrated acid. ("Do as you *oughta* — add acid to water.") Adding water to concentrated acid releases heat so violently it can erupt acid into your face. Bath make-up is usually authorized-personnel work — but every operator must know this rule.

CHAPTER 11

THE SAFETY PHILOSOPHY OF A CR(VI) LINE

THE MINDSET THAT TIES EVERY STATION TOGETHER — AND KEEPS YOU SAFE ACROSS A WHOLE CAREER

Rules and PPE are the *tools*. The philosophy is the *mindset* that makes them work. Seven principles guide everything on this line — more than the sulfuric manuals, because Cr(VI) earns it.

1. THE WORST HAZARD IS INVISIBLE — SO TRUST THE CONTROLS, NOT YOUR SENSES.

You can be over-exposed to chromic-acid mist without seeing or smelling it. The ventilation, the foam blanket, and the air monitoring exist *because* your nose can't be trusted here. Believe the instruments.

2. ENGINEERING CONTROLS COME BEFORE PPE.

Suppress the mist at the source, ventilate the tank, *then* wear your respirator. A respirator over broken ventilation is a false sense of safety.

3. HYGIENE IS A SAFETY CONTROL, NOT HOUSEKEEPING.

No eating, drinking, smoking, or face-touching; wash before breaks; keep work clothes separate. With a carcinogen, hand-to-mouth and take-home contamination are real exposure routes — your habits protect you and your family.

4. CONTAINMENT BEATS CLEANUP.

Move parts deliberately, let them drain over the tank, don't fling racks or splash. Most exposures aren't dramatic spills — they're drips and aerosols from sloppy handling. Slow, smooth movement is safer *and* makes better parts (less drag-out, less mist).

5. TWO CORROSIVES, MIRROR-IMAGE.

The **etch is caustic** (high pH); the **desmut and anodize are acid** (low pH). Both burn. Treat high pH and low pH with equal respect — and never let them mix outside a controlled neutralization.

6. HOT, ELECTRICAL, AND FLAMMABLE HAZARDS ARE STILL HERE TOO.

The bath is **warm and throws mist**; the seal can be **near boiling**; there's **high-current DC** at the work; the cathode bubbles **flammable hydrogen.** Ventilation on, ignition sources away, LOTO for electrical, dry hands.

7. WHEN IN DOUBT, STOP AND ASK.

The unsafe operator isn't the one who asks questions — it's the one who guesses. And **never let Cr(VI) reach a drain.**

★ **REMEMBER**

Cr(VI) mist first (invisible, carcinogenic — trust the controls); corrosives second (acid *and* caustic); thermal and electrical/gas third. If you can name the hazard family at the tank in front of you — and you know which tanks are Cr(VI) — you already know most of what you need to protect yourself.

★ **REMEMBER**

You now have the whole foundation: what anodizing is (and how it's *not* plating); the part-is-the-anode/grow-oxide idea; the **Cr(VI) hazard**; the thin gray paint-base oxide; minimal dimensional growth; the Type family and the move to BSAA/TSA; alloys; how the line works; PPE; emergency; and the safety mindset. Every station chapter builds on this. Turn the page knowing the *why*.

PART TWO — STATION BY STATION

STATION 01 OF 12

INSPECTION & RACKING

"IN ANODIZING, THE PART IS THE WIRE. A LOOSE PART IS A DEAD PART."

PURPOSE & THE "WHY"

Inspect the incoming aluminum part, confirm the **alloy and spec** (Type I or IB, thickness class, seal type, painted/natural), and **rack it** so it makes firm, current-carrying contact and hangs correctly. Because the part **is the anode**, all the current that grows your coating flows *through the rack contact into the part.* A loose or corroded contact means high resistance — **no coating, a thin coating, or a burned contact spot** — and possibly a dropped part in a chromic-acid tank (a Cr(VI) splash). Racking also decides *where* the un-anodized contact mark lands, how gas escapes, and how solution drains.

★ **REMEMBER — THE CONTACT MARK IS UNAVOIDABLE**

Wherever the rack grips, **no oxide grows** (that spot carries current, so it can't be insulated by its own coating). Place the contact where the mark won't matter — a hidden face, an edge, a hole — and make it **tight.**

ITEM	SPEC / PRACTICE	WHAT TO WATCH
Incoming inspection	Per traveler / print	Confirm **alloy/temper**, Type I or IB, thickness, seal, paint
Rack material	Titanium or aluminum	Ti racks don't anodize away; Al racks anodize and need stripping
Contact pressure	Firm, spring-loaded / bolted	Loose = burning, no coat, dropped parts (Cr(VI) splash)
Contact location	Hidden / non-critical area	The contact point won't anodize (bare mark)
Masking	Plugs/tape/lacquer on no-anodize areas	Threads, grounds, bearing fits, faying surfaces with other treatment
Orientation	Allow gas escape & drainage	Avoid trapped air (un-anodized voids); avoid solution pockets that drag out Cr(VI)

⚠ **HEADS-UP**

Racks and parts have **sharp edges and pinch points**, handled near a **carcinogenic acid tank.** A part that drops because of a weak rack becomes a **Cr(VI) splash hazard**, not just scrap. Rack tight, lift smooth.

💡 **TIP — TITANIUM RACKS ARE WORTH IT**

Titanium racks passivate rather than anodize, so they hold good contact load after load and don't need stripping. Aluminum racks anodize right along with the part — insulating the contact over time — so they must be **stripped/re-pointed** regularly.

STEP-BY-STEP

1. Read the traveler (alloy/temper, Type I/IB, thickness, masked areas, seal/paint). 2. Inspect for scratches/gouges/prior coatings — anodize *reveals* flaws, doesn't hide them. 3. Plan a hidden contact point that can carry full current. 4. Mask no-anodize areas per print. 5. Rack firmly — good metal-to-metal contact, correct orientation. 6. Verify no parts touch each other (shadowing). 7. Handle by the rack from here on.

SIGN-OFF — STATION 01

Teach-Back: Why contact matters more in anodizing than plating (part = anode); why the contact point won't anodize; why titanium racks are preferred; why orientation (gas/drainage) matters; two no-anodize areas that get masked and why; why a dropped part here is a Cr(VI) splash.

"I verified alloy and spec, made a firm contact in a non-critical area, masked all no-anodize features, racked for gas escape and drainage with no parts touching, and handled parts by the rack."

Trainee: ______________ Trainer: ______________ Date: __________

STATION 02 OF 12

CLEAN / DEGREASE

“ANODIZE OVER OIL AND YOU’VE GROWN A DEFECT, NOT A COATING.”

PURPOSE & THE “WHY”

Get the part **chemically clean** — no oil, grease, fingerprints, or shop soil — so the oxide grows evenly. Oxide can only grow uniformly from clean aluminum; oil and soil cause **uneven anodizing and streaks.** Cleaning is usually a **mild alkaline (non-etching) soak cleaner** — formulated *not* to attack the aluminum (we save controlled metal removal for the etch, and on Type I we keep even that light). The free, foolproof check is the **water-break test.**

💡 TIP — THE WATER-BREAK TEST (FREE, FAST, NEVER LIES)

Pull a cleaned part and watch the water. A truly clean surface sheets water in **one continuous, unbroken film** (PASS). A dirty surface makes water **bead up** (FAIL — re-clean). Costs nothing; takes two seconds; use it on every part.

```
FAIL - water beads up         PASS - water sheets evenly
+--------------+              +--------------+
|  o   o   o   |  X           |##############|  OK
|    o    o    |              |##############|
+--------------+              +--------------+
Oil remains -> RE-CLEAN       Surface clean -> proceed
```

PARAMETER	RANGE	WHAT TO WATCH
Type	Mild **non-etching alkaline** cleaner	Must not attack aluminum
Temperature	~50–70 °C (120–160 °F)	Per cleaner TDS
Time	~3–10 min	Heavy oil longer
Concentration / pH	Per TDS; mildly alkaline	Still a skin irritant/burn risk; titrate regularly
Water-break test	Pass = continuous sheet	The only reliable field check

⚠ HEADS-UP — HOT ALKALINE SOLUTION

Even a “mild” cleaner is hot and alkaline: splashes irritate/burn skin and eyes, and steam irritates airways. Splash goggles, gloves, apron. Skin → flush 15 min; eyes → eyewash 15 min and get help.

STEP-BY-STEP

1. Check temperature and concentration in range. 2. Immerse the racked load fully; start the timer; confirm agitation. 3. Withdraw slowly, draining back into the tank. 4. Run the **water-break test** — sheet → proceed; beading → re-clean. 5. Move promptly to the etch (don’t let a clean part sit and re-oxidize).

SIGN-OFF — STATION 02

Teach-Back: What the cleaner removes (oil/soil) vs. what it must *not* do (attack aluminum); read a water-break test; the caustic/irritant hazard and PPE; why a non-etching cleaner is used here.

“The part passed the water-break test, the cleaner was in range and at temperature, and I wore required PPE.”

Trainee: ____________________ Trainer: ____________________ Date: ____________

STATION 03 OF 12

ETCH (LIGHT)

“REMOVE THE OXIDE SKIN AND EVEN THE SURFACE — BUT ON TYPE I WE ETCH *LIGHT*, BECAUSE THE WHOLE POINT IS TO BARELY CHANGE THE PART.”

PURPOSE & THE “WHY”

Remove the natural oxide skin and a *thin, controlled* layer of aluminum in a **caustic etch** (or proprietary etch/deox), giving a uniform, active surface for anodizing. Aluminum always carries a thin, uneven natural oxide. The etch dissolves a small, controlled amount of aluminum to remove that skin and even out fine surface variation.

The Type I difference: because chromic anodize is chosen for **tight tolerance and fatigue life**, you keep the etch **light.** Heavy caustic etching removes metal, opens up surface defects, and can hurt fatigue performance — undoing the very reasons the part was specified chromic. Many aerospace flows use a **light etch or a mild deox** rather than a deep satin etch.

★ **REMEMBER — ETCH *LIGHT* ON TYPE I**

Unlike a decorative Type II part (where a heavy satin etch is fine), a Type I aerospace part wants the **minimum etch** needed to get a clean, active, uniform surface. Over-etching costs tolerance and fatigue life — the two things chromic exists to protect.

PARAMETER	RANGE	WHAT IT DOES / WATCH
Chemistry	**Sodium hydroxide** (light) or proprietary etch/deox	Removes oxide + a little aluminum
Concentration	Per TDS (often lighter than Type II)	Higher = faster/more aggressive
Temperature	~50–65 °C (120–150 °F)	Hot — burns and speeds etch
Time	**Short** (per spec)	Longer = more metal loss — avoid on Type I
Result	Uniform, active surface + **smut**	Smut is normal — removed at desmut

⚠ **HEADS-UP — HOT CAUSTIC IS ONE OF THE WORST BURNS ON THE LINE**

Caustic burns are **deep, get worse with time, and may not hurt much at first** — which makes people under-react. Full PPE: sealed goggles, face shield, elbow gloves, apron, boots. Skin → flush **15+ minutes**; eyes → eyewash 15 min and get medical help. Never assume “it’s just a little.”

⚠ **HEADS-UP — CAUSTIC + ALUMINUM MAKES HYDROGEN**

Caustic attacking aluminum releases **hydrogen gas** (flammable). Keep ignition sources away and ventilation running.

STEP-BY-STEP

1. Confirm concentration and temperature in range. 2. Immerse; start the timer for the **specified (short) etch time.** 3. Agitate gently per procedure. 4. Withdraw and drain back into the tank — the part will look **dark/smutty** (normal). 5. Move **directly to rinse** — don’t let caustic dry on the part.

SIGN-OFF — STATION 03

Teach-Back: What the etch removes (oxide + a little aluminum); **why Type I is etched *light* (tolerance + fatigue)**; why the part comes out smutty and what fixes it (desmut); the caustic-burn hazard and first response.

“I etched for the specified short time at the correct concentration/temperature, kept metal loss minimal to protect tolerance and fatigue, moved straight to rinse, and wore full PPE for hot caustic.”

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 04 OF 12

RINSE (THE FIREWALL)

“A RINSE ISN’T A REST STOP. IT’S A CHECKPOINT.”

PURPOSE & THE “WHY”

Wash the **caustic film** off the part before it reaches the acidic desmut and chromic anodize chemistry. The etch is caustic (high pH); desmut and anodize are acidic (low pH). Carry caustic forward and it **neutralizes and contaminates the acid baths**, wastes chemistry, and causes staining. The rinse knocks drag-out down by dilution. Cleanliness is tracked by **conductivity** (µS/cm) — *lower = cleaner.*

💡 **TIP — COUNTER-FLOW RINSING**

The professional approach is a **counter-current** cascade: parts move forward, water flows backward, so the cleanest water touches the part last. It saves water — important on a line with this many rinses.

PARAMETER	RANGE	NOTES
Temperature	`Ambient`	No heating needed
Time	`30–60 s with agitation`	Longer for cavities/blind holes
Water	`Municipal or DI`	DI preferred near anodize/seal
Conductivity	`< 500 µS/cm (target < 300)`	Monitor at shift start
Flow	`Continuous overflow / counter-current`	Prevents stagnation

⚠ **HEADS-UP**

This rinse carries dragged-in caustic and starts mildly alkaline. Goggles, gloves, non-slip footwear. **Wet floors and slips are the #1 everyday injury at rinses.**

DO

- Check conductivity at shift start; tell your lead if near/above limit.
- Transfer promptly from the etch — don’t let caustic dry.
- Immerse fully with agitation 30–60 s (longer for cavities).
- Lift and drain over the rinse tank before moving on.

TOP MISTAKES

- Treating the rinse as a quick dunk.
- Ignoring conductivity.
- Letting parts drip-dry between etch and rinse.
- Not flushing cavities.

SIGN-OFF — STATION 04

Teach-Back: Why carrying caustic into the acid stages causes problems; define drag-out/drag-in; what conductivity tells you and which way is “clean”; the conductivity limit/target.

“Rinse conductivity was within limit, water flow was running, and parts were rinsed the full time with PPE worn.”

Trainee: ____________ Trainer: ____________ Conductivity: ________ µS/cm

STATION 05 OF 12

DESMUT / DEOX

"ETCHING UNCOVERS THE ALLOY'S LEFTOVERS. DESMUT WASHES THEM AWAY — AND ON AEROSPACE ALLOYS THERE'S A LOT TO WASH."

PURPOSE & THE "WHY"

Remove the dark **smut** the etch left behind — the alloying elements (copper, silicon, iron, manganese) that didn't dissolve in caustic — using an **acid desmut/deox**, leaving a uniform, active surface for anodizing. On the **copper-rich 2xxx** and **zinc-rich 7xxx** aerospace alloys chromic is often run on, smut can be heavy, so a thorough desmut matters. Anodize over smut and you get **streaks, dull patches, and poor coating quality.** The chemistry depends on the alloy: simple alloys may need only **nitric** or **sulfuric**; high-copper/high-silicon alloys often need a **fluoride-bearing (or proprietary) deox.**

★ **REMEMBER**

Smut after etch is normal. Desmut's whole job is to remove it. A part that goes into the chromic tank still gray/smutty will come out streaky and dull — desmut is the uniform-surface gatekeeper.

PARAMETER	RANGE / PRACTICE	NOTES
Chemistry	Nitric and/or sulfuric; fluoride-bearing/proprietary deox for high-Cu/Si	Match to the alloy
Concentration	Per TDS	Titrate regularly
Temperature	Often ambient-warm	Per TDS
Time	~0.5–3 min	Until smut is gone, no longer
Result	Uniform, active surface	No gray film remaining

⚠ **HEADS-UP — STRONG ACID (AND POSSIBLY FLUORIDE)**

Desmut is acidic; **fluoride-bearing desmuts are especially hazardous** (hydrofluoric-type chemistry causes deep, delayed burns and is systemically toxic). Know exactly which chemistry your tank uses, read its SDS, wear full acid PPE, and know the **specific first-aid** (fluoride burns may require **calcium gluconate** per your SDS/medical plan). **Add acid to water** for any make-up.

STEP-BY-STEP

1. Confirm which desmut chemistry the tank is (and its hazards/first-aid). 2. Check concentration and temperature. 3. Immerse for the specified short time — just until smut clears. 4. Withdraw, drain, move to rinse. 5. Don't over-dip — excessive desmut can etch/pit some alloys and cost tolerance.

SIGN-OFF — STATION 05

Teach-Back: What smut is and where it came from; why anodizing over smut causes defects; that desmut chemistry depends on the alloy (fluoride for copper/silicon); the special hazard/first-aid of a fluoride desmut.

"I removed the smut with the correct desmut for the alloy, did not over-dip, and understand the specific hazard and first-aid for this tank."

Trainee: ____________ Trainer: ____________ Chemistry: ________

STATION 06 OF 12

RINSE (PRE-ANODIZE)

“THE LAST CLEAN HANDOFF BEFORE THE CARCINOGEN TANK.”

PURPOSE & THE “WHY”

Remove desmut acid before the part enters the chromic anodize tank. Carrying desmut drag-out into the anodize tank introduces unwanted ions — and, with a fluoride desmut, **fluoride contamination** that can roughen the coating. Just as important on a chromic line: a clean handoff keeps the **chromic-acid bath chemistry** in spec, and the cleaner the part goes in, the less you’ll drag *out* of the carcinogen tank later.

PARAMETER	RANGE	NOTES
Temperature	`Ambient`	—
Time	`30–60 s, agitation`	Longer for cavities
Water	`DI preferred`	Cleaner handoff to anodize
Conductivity	`Per shop spec`	Lower = cleaner
Flow	`Counter-current overflow`	—

⚠ **HEADS-UP**

Same rinse rules: drag-in acid, wet floors, goggles/gloves. If the prior tank was a **fluoride** desmut, this rinse water carries fluoride — handle and dispose accordingly.

💡 **TIP**

Move from this rinse into the anodize tank **promptly and consistently.** A long, variable dwell in air or rinse before anodizing can cause part-to-part coating differences. Consistency is quality.

STEP-BY-STEP

- Confirm flow and conductivity.
- Rinse the full time with agitation.
- Drain over the tank.
- Move straight to anodize.

TOP MISTAKES

- Quick dunk instead of a real rinse.
- Inconsistent dwell before anodize.
- Ignoring conductivity.
- Forgetting fluoride drag-in handling.

SIGN-OFF — STATION 06

Teach-Back: Why a clean handoff to the chromic tank matters (bath chemistry + less Cr(VI) drag-out later); what fluoride drag-in could do.

“The part was fully rinsed, conductivity was acceptable, and I moved promptly and consistently to the anodize tank.”

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 07 OF 12 · THE MAIN EVENT

ANODIZE — THE MAIN TANK

WHERE THE ALUMINUM'S OWN SURFACE BECOMES A THIN, GRAY, PROTECTIVE CERAMIC — AND WHERE YOU FACE THE CARCINOGEN HEAD-ON

Take a breath. Up to now, every station has had one job: get the part clean, even, and ready. **Station 07 is where the chromic anodize is created** — and where the **Cr(VI)** hazard is at its peak.

The big idea in one sentence: **we run the aluminum part as the *anode* in warm chromic acid and use a *ramped* DC voltage to grow a thin, dense, gray aluminum-oxide layer out of the part's own surface.** The cast:

- **Anode** — *your aluminum part* (**positive**). Where the oxide grows.
- **Cathode** — **steel or lead** plates (negative); in chromic anodizing a steel tank can itself be the cathode. They carry current and bubble off hydrogen; they do **not** feed metal into the part.
- **Electrolyte** — **chromic acid (CrO_3) ~30–100 g/L (~3–10%)**, held **warm (~32–40 °C / 90–105 °F)**, with **mist suppression, local exhaust, and a scrubber** running.

★ **REMEMBER — THE MIRROR IMAGE OF A PLATING TANK, *AND* THE CARCINOGEN OF THE LINE**

The part is the **anode** and **oxide grows out of it** (not deposited from the cathodes). And the bath is **hexavalent chromium.** Forget the first and the tank is a mystery; forget the second and it's a health hazard.

⚠ **HEADS-UP — CR(VI) CHROMIC ACID + WARM BATH + MIST (THE HEADLINE)**

Carcinogenic, corrosive, and **warm — which throws *more* mist than a cold sulfuric tank.** Engineering controls first (**foam/fume suppressant + local exhaust + scrubber**), respirator second, full PPE always. Any skin/eye contact → flush 15 min and get medical attention (chrome-ulcer / eye-corrosion risk). **Report any nosebleed or nasal soreness.** If suppression or exhaust is down, the tank is **not safe to run.**

• WHY TYPE I CHROMIC (AND WHAT MAKES IT PARTICULAR)

ADVANTAGE	WHAT IT MEANS ON THE FLOOR
Thin, fatigue-safe oxide	Won't significantly cut fatigue strength — chosen for stressed structure
Crevice/assembly-safe	Trapped residue far less corrosive than sulfuric
Minimal dimensional change	Good for tight-tolerance parts
Excellent paint/bond base	Primes and bonds beautifully
Corrosion protection (sealed)	Especially with a chromate seal

…but the same chemistry demands care: it's a **carcinogen** (mist control is non-negotiable), it runs on a **ramped voltage** (never slam to full voltage — you'll burn the part), it's sensitive to **chloride and sulfate contamination**, and the coating is **thin and gray** (don't expect bright color or wear resistance).

• THE BATH MAKE-UP — KNOW THE RANGES

COMPONENT / PARAMETER	TYPICAL RANGE	WHAT IT DOES / WATCH
Chromic acid (CrO_3)	`~30–100 g/L (~3–10%); often ~50 g/L`	The electrolyte; grows + gently dissolves the oxide
Temperature	`~32–40 °C (90–105 °F); often ~35 °C`	**Warmer than sulfuric** — gives the thin dense coat; also throws more mist
Voltage (DC)	`Ramped/stepped, not constant`	Classic cycles ramp to a **~22 V** hold (low-voltage/Type IB style) or step toward **~40 V** (conventional)
Current density	`Low; falls as oxide builds`	Much lower than sulfuric — chromic is a low-current process
Coating thickness	`~0.5–7.5 µm (often 2–5)`	Per spec; **thin** is the point
Cycle time	`~30–60 min (typical)`	Includes the voltage ramp + hold
Cathodes	`Steel or lead`	Carry current; bubble H_2; do not feed metal
Mist suppression	`Foam blanket / fume suppressant`	First line of defense against Cr(VI) mist
Ventilation	`Push-pull / lateral exhaust + scrubber`	Pull Cr(VI) mist off the surface; clean the exhaust
Chloride	`Very low (per spec, e.g., < ~0.2 g/L)`	Causes pitting/burning
Sulfate	`Very low (per spec, e.g., < ~0.5 g/L)`	Sulfate contamination harms the chromic coating — watch drag-in

★ REMEMBER — THE NUMBERS TO CARRY

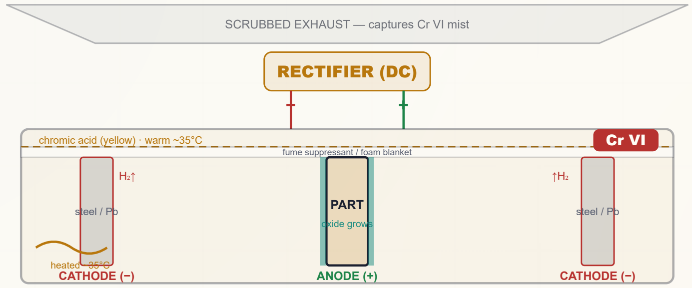
CrO_3 ~30–100 g/L (often 50), **temp ~32–40 °C (90–105 °F)**, **ramped voltage ~22→40 V**, **low current**, **~0.5–7.5 µm (often 2–5)**, **~30–60 min**, **steel/lead cathodes**, **mist suppression + exhaust + scrubber.** And above all: **this tank is Cr(VI).**

⚙ TECHNICAL STUFF — THE RAMPED/STEPPED VOLTAGE CYCLE (WHY YOU DON'T JUST SWITCH IT ON)

Chromic anodize uses a **gradually increasing voltage**, never an instant full-voltage start. A representative MIL-style cycle ramps from 0 V up to the hold voltage over several minutes, holds, and in some cycles steps higher near the end. Two common families: a **~22 V** low-voltage hold (associated with **Type IB**) and a conventional cycle stepping toward **~40 V** (the Bengough–Stuart lineage). Slamming on full voltage would concentrate current and **burn** the part; the slow ramp lets the insulating oxide build evenly. **Run the exact cycle on your spec/traveler.**

⚙ TECHNICAL STUFF — CONTAMINATION CONTROL

Chloride (from tap water, sweat, drag-in) causes **pitting/burning** even at low levels. **Sulfate** dragged in from sulfuric or a sulfuric desmut **degrades the chromic coating** — one reason rinses and bath segregation matter. Use DI water and keep contamination out; the lab controls the limits.



Key params: `CrO3 30-100 g/L · 32-40C (90-105F) · ramped 22->40 V · low CD · 0.5-7.5 um · foam + exhaust + scrubber · Cr VI`. Note the **reversed polarity vs. a plating cell:** the part is the ANODE (+).

• STANDARD OPERATING PROCEDURE

1. **Confirm the part is clean, lightly etched, desmutted, rinsed, and racked tight** — straight from the rinse, no long air dwell.
2. **Verify Cr(VI) controls FIRST:** foam/fume suppressant present and intact, push-pull exhaust running, scrubber on, your respirator and full PPE on.
3. **Verify the rest of the engineering controls:** bath at temperature (~32–40 °C), agitation per procedure, splash control in place.
4. **Check bath readiness:** CrO_3 in range (per lab), temperature in range, **chloride and sulfate within limits**, cathodes present/sound.
5. **Confirm rectifier polarity: PART = ANODE (POSITIVE).** *(Opposite of plating — verify every time.)*
6. **Set the voltage *ramp/step* program** for the part per the spec — **never start at full voltage.**
7. **Energize and ramp** per the cycle; watch voltage climb the ramp and the current behave as the insulating oxide grows.
8. **Run the full cycle**, monitoring temperature, voltage, current, agitation, **and the mist suppression/exhaust** the whole time.
9. **At cycle end, stop current, withdraw, drain over the tank** (let Cr(VI) drain back), and move to the post-anodize rinse — the pores are now open and ready for seal (or, rarely, dye).
10. **Log everything:** cycle, voltage/current, temperature, thickness checks, additions.

> ⚠ **HEADS-UP — VERIFY POLARITY EVERY SINGLE TIME**
> Everyone trained on plating expects the part to be the **cathode.** On this line **the part is the ANODE (+).** Reversed polarity gives no coating and can damage parts/cathodes. Check before you energize. And remember: Type I uses a **ramped** cycle — jumping straight to full voltage **burns** the part and can throw a Cr(VI) splash/mist.

COMMON DEFECTS — WHAT YOU'LL SEE AND WHAT IT MEANS

WHAT YOU SEE	MOST LIKELY CAUSE	WHAT TO DO
Powdery, soft, "burned" coating	Bath too warm; voltage ramped too fast/high; poor agitation	Check temp & cycle; verify agitation
Thin / no coating	Bad contact; wrong polarity; voltage cycle wrong; low CrO_3	Re-rack tight; **verify part = anode**; check cycle/chemistry
Pitting / burning	**Chloride contamination**; local current crowding	Lab-check chloride; use DI; improve racking
Degraded/streaky coating	**Sulfate contamination**; smut not removed; high-Cu/Si alloy	Lab-check sulfate; verify desmut; confirm alloy
Non-uniform on assembly	Trapped air / poor drainage in crevices	Re-orient for gas escape & drainage
Voltage won't follow the ramp / current odd	Poor contact; failing cathodes; bath out of spec	Check contacts/cathodes; lab-check bath

INTEGRATED SAFETY — THIS STATION DEMANDS TOTAL RESPECT

⚠ **HEADS-UP — CR(VI) CHROMIC ACID + MIST (THE HEADLINE)**

Carcinogenic, corrosive, warm. **Engineering controls first** (foam blanket + exhaust + scrubber), respirator second, full PPE always. Skin/eye contact → flush **15 minutes**, get medical attention (chrome-ulcer/eye risk). **Report nosebleeds/nasal soreness.** Acid to water for make-up.

⚠ **HEADS-UP — HIGH-CURRENT DC & FLAMMABLE HYDROGEN**

Shock/arc at contacts and bus bars — **LOTO** for all electrical work, dry hands only. Hydrogen off the cathodes is **flammable** — ventilation on; no flames, sparks, or grinding nearby.

⚠ **HEADS-UP — WATCH THE CONTROLS THE WHOLE CYCLE**

A run is 30–60 minutes; the warm tank throws mist the entire time. If the foam blanket dies (mist visible) or the exhaust stops pulling mid-run, you are being over-exposed. **Don't set-and-forget a carcinogen.**

TEACH-BACK — CAN YOU ANSWER?

- Which electrode is your part, and which charge? *(Anode, +.)*
- Where does the coating come from? *(The part's own Al + O_2.)*
- The headline numbers (CrO_3, temp, ramped V, thickness, time)?
- Why is the bath warm and why does that raise the mist hazard?
- What does chloride do? What does sulfate do?
- What are the **five pillars** of Cr(VI) control?

TOP MISTAKES TO AVOID

- Assuming the part is the cathode (it's the **anode**).
- **Slamming full voltage instead of ramping.**
- Running with dead mist suppression or failed exhaust.
- Loose contact; ignoring chloride/sulfate.
- Eating/face-touching near the tank.

SIGN-OFF — STATION 07

"I can independently: verify mist suppression, exhaust, scrubber, agitation, temperature, and PPE/respirator before running; verify bath readiness (CrO_3, temp, chloride, sulfate per lab); confirm the part is racked as the ANODE (positive); run the correct ramped voltage cycle without burning the part; recognize burning/pitting/thin-coat/sulfate defects; and state the Cr(VI), acid, electrical, and hydrogen hazards plus the five pillars of Cr(VI) control. I understand chemistry adjustments are made only by authorized personnel, and that Cr(VI) never goes to a drain."

Operator: ____________________ Date: __________ Trainer: ____________________ Date: __________

STATION 08 OF 12

RINSE (POST-ANODIZE)

“GENTLE NOW — THE PORES ARE OPEN, THE PART IS HOT, AND THE WATER IS FULL OF CARCINOGEN.”

PURPOSE & THE “WHY”

Rinse the chromic acid out of the freshly anodized part **without contaminating or prematurely sealing the open pores** — and **capture the dragged-out Cr(VI)** so it doesn’t spread or reach a drain. The fresh oxide is **porous and absorbent**; leftover chromic acid interferes with sealing (and any rare dye). Use **clean DI** and keep it **cool** before any dye; if going straight to seal, follow your spec. On a chromic line this rinse is also a **drag-out recovery** point — its water is Cr(VI)-contaminated and routes to **treatment**, never to drain.

★ **REMEMBER — DON’T PRE-SEAL BY ACCIDENT, AND DON’T LOSE THE CHROMIUM**

Use **cool/ambient DI** before any dye (hot water here can start closing pores). And treat this rinse water as **Cr(VI) waste** — it goes to reduce-then-precipitate treatment (Part Three), never down a drain.

PARAMETER	RANGE	NOTES
Water	`DI strongly preferred`	Pores absorb impurities
Temperature	`Cool / ambient (before dye)`	Avoid premature sealing
Time	`1–3 min, gentle agitation`	Flush chromic acid from pores
Conductivity	`Low (per spec)`	Confirms acid removed
Waste routing	**`To Cr(VI) treatment`**	Chromium-bearing — never to drain

⚠ **HEADS-UP**

This rinse carries dragged-out **chromic acid (Cr VI)** — corrosive *and* carcinogenic, plus a slip hazard. Gloves/goggles; smooth handling to minimize aerosols. Its water is **regulated waste.**

STEP-BY-STEP

- Confirm DI flow, cool temperature.
- Rinse gently, full time.
- Drain over the tank.
- To seal (or, in the rare colored job, to dye).

TOP MISTAKES

- Hot water before dye (pre-seals).
- Hard tap water (impurities).
- Short rinse leaving acid behind.
- Letting Cr(VI) rinse water reach a drain.

SIGN-OFF — STATION 08

Teach-Back: Why the pores are absorbent now; why cool DI before any dye; **why this rinse water is Cr(VI) waste.**

“I rinsed with cool DI to flush chromic acid from the open pores without pre-sealing, handled it as Cr(VI)-contaminated water, and moved promptly to seal.”

Trainee: ____________________ Trainer: ____________________ Date: _____________

STATION 09 OF 12 · USUALLY SKIPPED ON TYPE I

DYE (USUALLY NOT USED)

“TYPE I IS THIN AND GRAY, SO IT’S RARELY DYED — BUT HERE’S HOW THE STEP WORKS, AND WHY YOUR PART PROBABLY SKIPS IT.”

PURPOSE & THE “WHY”

On dyeable anodize, this step draws **color into the open pores.** On **Type I**, the coating is **thin and gray/opaque**, so it carries color poorly and is **usually left natural or sent to primer/paint instead.** Most chromic parts go **anodize → rinse → seal**, skipping dye. We keep the station to parallel the rest of the family. *If* a colored chromic part is specified (rare, usually a dark shade), dye goes in **before** sealing, into open pores, in DI water, with consistent parameters — then the part **must be sealed.**

★ **REMEMBER — FOR TYPE I, DYE IS THE EXCEPTION**

Don’t expect a bright color on a thin gray chromic coat. If your traveler doesn’t call a dye color, **skip this station and go to seal.** If it does, **dye before seal, never after** — sealing closes the pores.

PARAMETER (IF DYED)	RANGE / PRACTICE	NOTES
Dye type	Organic	Muted on thin gray coat; darks only, realistically
Temperature	`~50–60 °C typical`	Affects uptake
pH / time / water	`Per dye TDS; DI`	Consistency controls shade
Coating thickness	`Adequate for shade`	Type I is thin — limits depth of color

⚠ **HEADS-UP — DYE HANDLING**

Some dyes are **skin/eye irritants or sensitizers**; baths may be warm/mildly acidic. Gloves, goggles, read each dye’s SDS (some carry disposal requirements).

💡 **TIP**

If a customer wants a *colored* aerospace part, ask whether **Type II** (which dyes beautifully) is acceptable — or whether the color is meant to be **paint over Type I.** Don’t promise bright dyed color on chromic.

STEP-BY-STEP (IF DYED)

- Confirm dye type/temp/pH (DI).
- Part from a cool DI rinse (pores open).
- Immerse fully, agitate to target shade.
- Drain, rinse off surface dye; move promptly to seal.

TOP MISTAKES

- Sealing before dyeing.
- Expecting bright color on thin gray coat.
- Inconsistent parameters.
- Not rinsing surface dye.

SIGN-OFF — STATION 09

Teach-Back: Why Type I is usually *not* dyed (thin/gray, paint base); the correct order if it is (anodize → rinse → dye → rinse → seal); dye-handling hazards.

“I confirmed whether a dye color was specified; for natural/painted parts I skipped dye and went to seal; if dyed, I dyed before sealing into open pores in DI.”

Trainee: ____________________ Trainer: ____________________ Color or “N/A”: __________

STATION 10 OF 12

SEAL (CHROMATE OR HOT WATER)

"THE FINAL LOCK — CLOSE THE PORES TO KEEP CORROSION OUT. ON A CHROMATE SEAL, YOU'RE BACK IN CR(VI) TERRITORY."

PURPOSE & THE "WHY"

Close the open pores of the anodic oxide so the coating reaches its full corrosion resistance (and any rare dye is locked in). An unsealed anodize is porous — it can absorb stains and corrode through the pores. The two common Type I seals:

- **Dilute chromate (sodium dichromate) seal** — a hot, dilute dichromate solution. It closes the pores **and** deposits **corrosion-inhibiting chromate** into the coating, boosting salt-spray life. **This seal is itself hexavalent chromium — same carcinogen, same controls, same waste handling.**
- **Hot DI water seal (~96–100 °C)** — near-boiling water converts the pore-wall oxide to **boehmite**, which swells the pores shut. **Metal-free and chromium-free** — preferred where Cr(VI) must be minimized.

★ **REMEMBER — SEALING IS THE POINT OF NO RETURN**

After sealing you **cannot dye.** A poorly sealed part corrodes; an over-sealed/contaminated part can show **"sealing bloom"** (a chalky film). And note: a **dichromate seal is Cr(VI)** — don't drop your guard at the seal tank.

THE TWO COMMON TYPE I SEALS

SEAL TYPE	TEMPERATURE	TRADEOFFS
Dilute sodium-dichromate (chromate)	Hot (per TDS)	Best corrosion protection; deposits inhibitor — **but is Cr(VI)** (mist, skin, waste controls apply)
Hot DI water	~96–100 °C	Simple, **chromium-free**; high energy, slow; prone to **bloom** if water/pH off

⚙ **TECHNICAL STUFF — WHY EACH SEAL WORKS**

Hot water hydrates the pore-wall oxide to **boehmite (AlOOH)**, which occupies more volume and **swells the pore mouths shut.** A dichromate seal both closes the pores and leaves **chromate** in the coating that actively *inhibits* corrosion (it self-heals minor scratches) — which is why it gives the best salt-spray numbers and why aerospace specs historically called for it. Both reach the same goal — closed, protected pores — by different routes.

⚙ **TECHNICAL STUFF — "SEALING BLOOM"**

A powdery white/gray film after sealing usually comes from **hard water, drag-in, wrong pH, or over-concentrated seal.** The real fix is **clean DI water and correct seal chemistry/pH/temperature.**

⚠ HEADS-UP — THE CHROMATE SEAL IS *ALSO* CR(VI)

A sodium-dichromate seal carries the **same carcinogen hazards** as the anodize bath — mist (it's hot), skin/eye corrosion, and **regulated waste.** Use mist control/ventilation, full PPE/respirator per program, and route its waste to **Cr(VI) treatment.** A **hot-water seal** avoids this — one big reason shops choose it.

⚠ HEADS-UP — NEAR-BOILING TANK = SCALD HAZARD

A hot-water or hot chromate seal runs hot to near-boiling. Splashes and steam cause **serious scald burns.** Face shield, heat-resistant gloves, long sleeves; mind the steam; lower parts gently.

MAKE-UP & KEY PARAMETERS

PARAMETER	RANGE / PRACTICE	NOTES
Seal type	Dilute dichromate **(Cr VI)** or hot DI water	Per spec
Temperature	`Per chosen seal (hot water ~96–100 °C)`	Wrong temp = poor seal
pH	`Per seal product`	Wrong pH = poor seal / bloom
Time	`Per seal type & coating thickness`	—
Result	Closed pores; passes seal/corrosion test	—

STEP-BY-STEP

1. Confirm seal type, temperature, pH, water. 2. Confirm part came from a clean rinse (no contamination; surface dye rinsed if dyed). 3. Immerse gently (mind the scald), full time. 4. Withdraw, rinse if specified, inspect for **bloom.** 5. Move to final rinse/dry. *(If chromate seal: same Cr(VI) controls and waste routing as the anodize tank.)*

TEACH-BACK — CAN YOU ANSWER?

- What sealing does (closes pores) and why.
- Why you can't dye after sealing.
- The two seal types and one tradeoff of each.
- **That a dichromate seal is Cr(VI).**
- The scald hazard; what sealing bloom is.

TOP MISTAKES

- Trying to dye after sealing.
- Wrong temp/pH (incomplete seal).
- Hard/tap water (bloom).
- Plunging into a near-boiling tank.
- **Treating a dichromate seal as harmless** / routing its waste to drain.

SIGN-OFF — STATION 10

"I sealed at the correct type/temperature/pH after anodizing, checked for bloom, handled the scald hazard, and — if a chromate seal — applied full Cr(VI) controls and waste routing."

Trainee: ____________________ Trainer: ____________________ Seal type & temp: ____________

STATION 11 OF 12

RINSE / DRY

"FINISH CLEAN, DRY GENTLY, DON'T UNDO YOUR WORK — AND THE RINSE WATER MAY STILL BE CHROMIUM-BEARING."

PURPOSE & THE "WHY"

Give the sealed part a final clean rinse and a **gentle dry** without water spotting or thermal damage. A clean final rinse removes seal-tank drag-out so the part dries spot-free. If the seal was **chromate**, this rinse water is **chromium-bearing** and routes to treatment.

PARAMETER	RANGE	NOTES
Final rinse water	`DI`	Spot-free drying
Rinse temperature	`Warm`	Aids drying
Dry method	`Air / warm forced air; gentle oven`	Avoid excessive heat
Handling	`By rack / clean gloves`	No bare-hand prints on finished parts
Waste (if chromate seal)	`To Cr(VI) treatment`	Chromium-bearing rinse

⚠ **HEADS-UP**

Parts coming off a hot seal are **hot** — burn risk. Let them cool or use gloves. If the seal was chromate, **treat the rinse water as Cr(VI) waste.**

💡 **TIP**

A **DI final rinse** is the cheapest insurance against water spots on a finished part — and if the part is headed for **primer/paint**, a clean, spot-free, correctly-dried surface is essential for good adhesion.

STEP-BY-STEP

- Final DI rinse.
- Drain.
- Gentle dry.
- Handle by rack/gloves to inspection.

TOP MISTAKES

- Tap-water final rinse (spots).
- Over-aggressive drying.
- Bare-hand handling.
- Sending chromate rinse to drain.

SIGN-OFF — STATION 11

Teach-Back: Why a clean final rinse + gentle dry protects the job and the paint base; chromate rinse waste routing.

"I gave a clean DI final rinse, dried gently without spotting/overheating, handled parts cleanly, and routed any chromium-bearing rinse to treatment."

Trainee: ____________________ Trainer: ____________________ Date: ______________

STATION 12 OF 12

FINAL INSPECTION

“PROVE IT’S RIGHT — THICKNESS, SEAL/CORROSION, APPEARANCE, AND DIMENSIONS.”

PURPOSE & THE “WHY”

Verify the finished part meets spec: **coating thickness, seal/corrosion quality, appearance, and dimensions.** Type I quality isn’t visible by eye alone. Because chromic coatings are **thin and gray**, thickness measurement can be harder and the acceptance is often by **corrosion performance and appearance** plus spec-defined checks.

CHECK	METHOD	PASS CRITERIA
Coating thickness	Eddy-current (ASTM B244) where measurable; micro-section if needed	Within spec (thin — e.g., 0.5–7.5 µm per print)
Seal / corrosion	**Salt spray (ASTM B117)** per spec; dye-stain/admittance where applicable	Meets spec corrosion life
Appearance	Visual under good light	Uniform gray; no burns, bloom, pits, streaks, bad contact marks
Dimensions	Gauging where critical	Within tolerance (growth tiny but confirm critical fits)
Paint/bond readiness	Per customer process	Clean, correctly sealed surface for primer/adhesive

★ **REMEMBER — FOR TYPE I, CORROSION AND APPEARANCE LEAD**

Because the coating is thin and gray, the headline acceptance is usually **corrosion performance (salt spray) and a uniform, defect-free appearance** at the spec’d thickness — not color. A part can be the right gray and *still fail* if it isn’t properly sealed or it pitted.

💡 **TIP — MEASURING A THIN GRAY COAT**

Eddy-current readings on very thin chromic coatings can scatter; take multiple readings on **significant surfaces** and follow your spec’s method. When in doubt, a **micro-section** is the referee.

STEP-BY-STEP

- Thickness on significant surfaces.
- Seal/corrosion test per spec.
- Visual for defects; gauge critical dimensions.
- Confirm paint/bond readiness.
- Accept / reject / route to rework (strip & re-anodize).

TOP MISTAKES

- Measuring only easy surfaces.
- Skipping the corrosion/seal check.
- Judging under bad light.
- Assuming “thin = no dimensional check.”

SIGN-OFF — STATION 12

Teach-Back: Which checks lead for Type I (corrosion + appearance at spec thickness); why a thin gray coat still needs a seal/corrosion test.

“I verified thickness on significant surfaces, ran the seal/corrosion test per spec, inspected for defects, confirmed critical dimensions, and confirmed paint/bond readiness.”

Trainee: ________________ Trainer: ________________ Thickness ______ µm · Seal/Corrosion: P/F

PART THREE — ACROSS THE LINE
THE CROSS-CUTTING TOPICS

ACROSS THE LINE

THE IDEAS THAT DON'T LIVE IN ONE TANK — THEY TIE THE WHOLE LINE TOGETHER

• COATING THICKNESS & DIMENSIONAL GROWTH (DEEP DIVE)

Thickness is set by the voltage cycle and time at the bath temperature — not by a simple current-density × time rule the way sulfuric is, because chromic runs a *ramped* voltage and a low, falling current. Run the **spec'd cycle**; verify with **eddy-current** on the **significant surface** (or micro-section for very thin coats). **Dimensional growth** follows the universal half-in/half-out rule, but because Type I is **thin (~0.5–7.5 µm)**, the change is tiny — each surface moves out ~half the thickness, a diameter changes by ~the full thickness. Plan tolerances/masking only on truly critical features.

> ★ **REMEMBER**
> Type I thickness comes from **the spec'd voltage cycle + time** (verify by eddy-current/micro-section), and growth is **small but not zero** — confirm critical fits.

• THE FATIGUE-STRENGTH ADVANTAGE (DEEP DIVE) — *WHY AEROSPACE CHOSE CHROMIC*

Anodizing removes and converts surface metal, and the resulting oxide and surface condition can act as **fatigue-crack initiation sites** — a real penalty with thick **sulfuric** coatings on stressed parts. **Chromic anodize causes far less fatigue debit** because it's **thin** and grows in a **gentle** electrolyte that removes little base metal. On **fatigue-critical structure** (spars, fittings, gear parts) that difference can be the deciding factor. This is also *why the etch is kept light* (Station 03): heavy metal removal during prep would give back the fatigue advantage you anodized chromic to keep.

> ★ **REMEMBER**
> **Chromic = low fatigue debit.** Thin coating + gentle acid + light etch = the fatigue-friendly finish. Over-etch the part and you throw that advantage away.

• THE CREVICE / ASSEMBLY / FAYING-SURFACE ADVANTAGE (DEEP DIVE)

On built-up assemblies — **lap joints, spot-welds, riveted structure, blind holes, faying (mating) surfaces** — some electrolyte inevitably gets **trapped** in the crevices. **Trapped sulfuric** keeps attacking the aluminum and drives corrosion from the inside. **Trapped chromic** residue is **far less aggressive** (and the chromate left behind even *inhibits* corrosion). That's why chromic is preferred for **assemblies and faying surfaces** that can't be perfectly rinsed.

> ★ **REMEMBER**
> In a crevice you can't fully rinse, **chromic residue is the safer residue.** That single fact is why assemblies and faying surfaces are chromic-anodized.

• PAINT-BASE & ADHESIVE-BOND USE (DEEP DIVE)

Type I's biggest commercial role isn't color — it's **adhesion.** The thin, porous, chemically-active oxide is one of the best **primer/paint and structural-adhesive bases** in the family. For painted parts, the spec may call a **lighter seal** (or a specific chromate seal) so the surface keeps the right "tooth" for the primer. For **adhesive bonding**, a related family (e.g., **phosphoric-acid anodize, PAA**) is sometimes specified instead — but chromic remains a workhorse paint base. **Match the seal to the downstream coating per the spec.**

> ★ **REMEMBER**
> **Type I's job is grip, not color.** Seal type is chosen to suit the **paint/primer/adhesive** that follows — confirm it on the traveler.

• RACKING & ELECTRICAL CONTACT (DEEP DIVE)

The part is the anode, so **the rack is the wire that grows your coating.** Firm current-carrying contact; contact placed where the bare mark won't matter; orientation for **gas escape and drainage** (no trapped air = no un-anodized voids; good drainage = less Cr(VI) drag-out); no parts touching; maintained racks (titanium preferred). Poor contact is the most common cause of thin coatings, burned spots, and dropped parts. *(See Station 01.)*

• SEALING (DEEP DIVE)

Two routes: **dilute dichromate** (best corrosion, deposits inhibitor — **but is Cr VI**) and **hot DI water** (chromium-free, simple, bloom-prone if water/pH off). Universal enemies: **hard water, wrong pH, contamination → bloom.** Confirm the seal/corrosion result per spec (salt spray; dye-stain/admittance where applicable).

> ⚠ **HEADS-UP**
> A **dichromate seal is Cr(VI)** — mist, skin, and waste controls apply exactly as at the anodize tank, and its rinse/spent solution goes to **Cr(VI) treatment, never to drain.** Hot-water seal is the chromium-free alternative.

• ALUMINUM ALLOY EFFECTS (DEEP DIVE)

Chromic is often run on **copper-rich 2xxx (2024)** and **zinc-rich 7xxx (7075)** aerospace alloys — duller, heavier smut, but the application accepts "sound and gray." Rack like-alloy with like-alloy; anodize a sample of the **actual alloy/lot** when appearance/corrosion is critical. *(See Chapter 7.)*

• RINSING & WATER/DI MANAGEMENT

Rinses are firewalls (caustic etch vs. acid desmut/anodize) **and** Cr(VI) drag-out recovery points downstream of Station 07. Use **counter-current** cascades and **DI** near the anodize/seal. Track **conductivity** (lower = cleaner). Keep **chloride and sulfate** out of the chromic bath. **All chromium-bearing rinse goes to treatment — never to drain.**

★ **REMEMBER**

Downstream of the anodize tank, *every* rinse is doing two jobs at once: keeping chemistries apart **and** corralling carcinogenic chromium. A sloppy rinse isn't just a quality miss — it's a containment failure.

⚙ **TECHNICAL STUFF — CHLORIDE VS. SULFATE, THE TWO CONTAMINANTS TO FEAR**

Chloride causes **pitting and burning** of the coating even at low levels — it sneaks in from tap water and hand sweat, so use DI. **Sulfate** dragged in from sulfuric chemistry **degrades the chromic coating** — one more reason chromic lines segregate their baths and rinses from sulfuric work. The lab sets and checks both limits.

☀ **FUN FACT**

Anodizing's reliance on clean water is why many shops run their whole back half — post-anodize rinse, seal, final rinse — entirely on deionized water. The same DI that prevents water spots also keeps the chloride out that would pit the coating.

RENDERING THE CARCINOGEN SAFE

CR(VI) WASTE TREATMENT

WHERE THE CARCINOGEN IS RENDERED SAFE — CHEMISTRY EVERY OPERATOR SHOULD UNDERSTAND

A chromic anodize shop can't pour spent chemistry or rinse water down the drain. Cr(VI) must be **treated** to make it safe, the air must be **controlled and permitted**, and the chromium waste must be **handled as hazardous.**

• THE CORE REACTION: REDUCE, THEN PRECIPITATE

The standard wastewater treatment for hexavalent chromium is a **two-step:**

1. **Reduce $Cr^{6+} \rightarrow Cr^{3+}$.** Add a **reducing agent** (commonly **sodium metabisulfite, sulfur dioxide, or ferrous sulfate**) at **low pH (acidic).** This converts the dangerous, soluble hexavalent chromium into the far-less-hazardous **trivalent** chromium. (You can often *see* the color shift from yellow toward green.)
2. **Precipitate as hydroxide and remove. Raise the pH** (add caustic/lime) so the trivalent chromium drops out as insoluble **chromium hydroxide sludge**, which is **settled/filtered** out. Treated water is tested before discharge; the chromium sludge is managed as **hazardous waste.**

★ **REMEMBER**

Reduce then precipitate: $Cr^{6+} \rightarrow Cr^{3+}$ (acidic + reducing agent), then raise pH to drop out the chromium as sludge and filter it. This is the single most important environmental concept on a chromic line.

⚙ **TECHNICAL STUFF**

Reduction needs **acidic** conditions to go fast and complete; precipitation needs **alkaline** conditions. So treatment swings pH down (reduce) then up (precipitate) — and the final discharge pH and chromium level must meet permit limits before any water leaves. The trivalent-hydroxide sludge is a regulated **hazardous waste** with manifesting and disposal requirements.

• AIR PERMITS & MIST CONTROL

Because chromic-acid mist is a regulated carcinogen, Type I tanks are controlled at the source (**fume suppressants/foam, push-pull/lateral ventilation**) and the exhausted air is usually cleaned by a **fume scrubber** before release. The operation typically requires an **air permit**, and there are federal emission rules specifically for chromium anodizing/plating. Your shop's permit dictates the controls — keeping the foam blanket and exhaust working is part of staying compliant **and** keeping you safe.

⚠ **HEADS-UP**

These controls protect **you** (the mist you don't breathe) *and* the **public/environment** (the chromium that doesn't leave the stack or the drain). A dead foam blanket or a bypassed scrubber is **both** a personal exposure and a permit violation. Treat them as non-negotiable.

☀ **FUN FACT**

The yellow-to-green color change during reduction is chemistry your eyes can track in the treatment tank: yellow = hexavalent (dangerous), green = trivalent (much safer). The plant is literally turning the carcinogen into something far less harmful, one color shift at a time.

HOW WE PROVE A PART IS GOOD

QUALITY CHECKS & TEST METHODS

THE INSTRUMENTS AND STANDARDS THAT TURN “LOOKS FINE” INTO “MEETS SPEC”

CHECK	METHOD (TYPICAL STD)	ANSWERS
Coating thickness	Eddy-current (ASTM B244); micro-section for thin coats	“Enough oxide grown?”
Corrosion / seal	**Salt spray (ASTM B117)** per spec; dye-stain (B136)/admittance (B457) where applicable	“Will it survive service? Pores closed?”
Appearance/defects	Visual under controlled light	Burns, bloom, pits, streaks, marks, uniform gray
Paint/bond adhesion	Per customer process	“Good base for primer/adhesive?”
Fatigue (when required)	Per spec coupon testing	“Fatigue debit acceptable?”
Dimensions	Gauging where critical	Within tolerance after (small) growth

★

REMEMBER — THE TWO ACCEPTANCE LEADERS ON TYPE I

Corrosion (salt spray) = “will it survive and are the pores sealed?” and **appearance = “uniform gray, no defects?”** Thickness still matters and is verified, but on a thin gray chromic coat the headline acceptance is usually corrosion + appearance at the spec thickness — not color.

💡

TIP — THE SEAL/CORROSION CHECK IS NON-NEGOTIABLE

A chromic part can be the right gray and the right thickness and *still fail* if it isn’t sealed or it pitted. That’s exactly why the salt-spray (and, where applicable, dye-stain/admittance) seal test exists — it catches the failure your eye can’t see.

⚙

TECHNICAL STUFF — WHY THIN COATS NEED THE MICRO-SECTION REFEREE

Eddy-current gauges read coating thickness by how the oxide spaces the probe off the conductive aluminum. On a very thin chromic coat the signal is small and scatters, so readings vary. When a number is borderline or disputed, a destructive **micro-section** (cut, mount, polish, measure under a microscope) is the referee method written into many specs.

FIELD REFERENCE

TROUBLESHOOTING QUICK-INDEX

FIND YOUR SYMPTOM, READ THE LIKELY CAUSE, TRY THE FIX

DEFECT / SYMPTOM	LIKELY CAUSE	FIX
Burning (powdery, dark, rough)	Bath too warm; voltage ramped too fast/high; poor agitation	Verify temp & **ramp/step program**; restore agitation
Thin / no coating	Poor contact; **wrong polarity (part not anode)**; wrong/low voltage cycle; low CrO_3	Re-rack tight; **verify part = anode**; run spec cycle; lab-check bath
Pitting	**Chloride contamination**; current crowding	Lab-check chloride; switch to DI; improve racking
Degraded/streaky coating	**Sulfate contamination**; smut left on; high-Cu/Si alloy	Lab-check sulfate; verify desmut; confirm alloy
Non-uniform on assemblies	Trapped air/poor drainage in crevices	Re-orient for gas escape & drainage
Sealing bloom (chalky film)	Hard water; wrong pH; over-concentrated seal	DI water; correct pH/concentration; light post-dip
Poor corrosion (fails salt spray)	Incomplete seal; chromate seal under-strength	Correct seal temp/pH; re-seal; verify chromate make-up
Contact burns / no-coat at contact	Loose/insufficient rack contact	Tighten/add contacts; re-point/replace rack
Visible mist over the tank	**Foam blanket dead / exhaust failing**	**STOP — this is a Cr(VI) over-exposure; restore suppression/exhaust before running**
High Cr in rinse/discharge	Treatment incomplete / drag-out to drain	Verify reduce-then-precipitate; never discharge untreated

💡 **TIP**

Most chromic-line defects trace to one of five roots: **temperature/voltage cycle** (burning), **contamination** (chloride → pitting; sulfate → degraded coat; hard water → bloom), **alloy** (dull/gray — often acceptable), **order/handling** (contact, dwell, seal), or **controls** (mist/exhaust). Name the root, not just the symptom.

★ **REMEMBER**

Several rows here are **safety-relevant**, not just quality: **wrong polarity** (verify part = anode), **visible mist** (a Cr(VI) over-exposure — stop), and **Cr in discharge** (a reportable release). When in doubt about a bath change, ask your supervisor before you act.

EVERY TERM, PLAIN AND SIMPLE

GLOSSARY

PLAIN-LANGUAGE DEFINITIONS FOR EVERYTHING IN THIS MANUAL

Action Level: 2.5 μg/m³ (8-hr avg) Cr(VI) trigger for monitoring/medical surveillance under OSHA 29 CFR 1910.1026.
Admittance test: electrical test (ASTM B457) of how "open" the coating is — used to confirm sealing.
Anode: the **positive** electrode. In anodizing it's **your part** — where the oxide grows.
Anodizing: growing an integral oxide layer out of a metal's surface by making the part the anode in an electrolyte.
Barrier layer: the thin dense oxide at the bottom of the coating, against the metal.
Bengough–Stuart process: the classic 1920s chromic anodize voltage cycle that Type I descends from.
Boehmite: hydrated aluminum oxide (AlOOH) formed during hot-water sealing; its swelling closes pores.
BSAA (Boric-Sulfuric Acid Anodize): the leading **non-chromic substitute** for Type I; the common **Type IC** process.
Cathode: the **negative** electrode (steel or lead here) — carries current and bubbles hydrogen; does **not** feed the coating.
Chromate seal: a dilute sodium-dichromate seal that closes pores and deposits corrosion inhibitor — **and is itself Cr(VI).**
Chromic acid: chromium trioxide (CrO_3) in water — the Type I electrolyte; **hexavalent chromium, a carcinogen.**
Cr(VI) / hexavalent chromium: the +6 chromium in chromic acid; a **confirmed human carcinogen.**
Cr(III) / trivalent chromium: the far-less-hazardous reduced form; the goal of waste treatment.
Dimensional growth: the coating grows **~half in, half out**; tiny on thin Type I.
Drag-out / drag-in: solution clinging to a part leaving a tank (out) / contaminating the next tank (in) — on this line it also spreads Cr(VI).
Eddy-current gauge: non-destructive thickness instrument for coatings on aluminum.
Etch: a (light, on Type I) caustic dip removing the natural oxide and a little aluminum.
Faying surface: a mating surface in an assembly — chromic is preferred here (gentle residue).
Fume suppressant / foam blanket: surfactant additive that forms foam and cuts Cr(VI) mist at the source.
MIL-PRF-8625 (MIL-A-8625): the spec defining anodize **Type I (chromic), IB (low-voltage chromic), IC (non-chromic), II (sulfuric), III (hardcoat).**
PEL: 5 μg/m³ (8-hr) Cr(VI) Permissible Exposure Limit (OSHA 1910.1026).
Porous layer: the thick upper oxide full of vertical pores; thin on Type I.
Reduce-then-precipitate: $Cr^{6+} \rightarrow Cr^{3+}$ (acidic + reducing agent), then raise pH to drop out chromium sludge — the waste-treatment method.
Scrubber: device that washes Cr(VI) out of exhausted air before release.
Sealing: the final step that **closes the pores** (chromate or hot water).
Significant surface: the surface that must meet thickness/finish spec; where you measure.
Smut: dark residue of alloying elements (Cu, Si, Fe) left by caustic etching; removed at desmut.
TSA (Thin-film Sulfuric Anodize): another **chromic-free substitute** for thin protective coatings/paint base.
Type I: the **chromic** anodize — thin, gray, aerospace/fatigue/paint-base, **Cr VI.**
Valve metals: metals that anodize to protective oxides (aluminum, titanium, magnesium, tantalum, niobium).
Water-break test: free cleanliness check — clean metal sheets water; dirty metal beads it.

> ★ **REMEMBER**
> Of all these, five matter most: **anode** (your part), **Cr(VI)** (the carcinogen bath — invisible mist), **thin/fatigue-safe/paint-base** (Type I's whole point), **ramped voltage** (never slam it on), and **reduce-then-precipitate** (never let Cr(VI) reach a drain).

TEMPLATE — PRINT ONE PER OPERATOR

OPERATOR TRAINING MATRIX

HOW A SUPERVISOR PROVES — ON PAPER — THAT EACH OPERATOR IS TRAINED AND SIGNED OFF BEFORE RUNNING SOLO

Sign-off levels: O = Observed only · **A** = Assisted under supervision · **Q** = Qualified to run solo · **T** = Qualified to Train others.

#	STATION / SKILL	O/A/Q/T	TRAINER INIT & DATE	OPERATOR INIT & DATE	RE-CERT DUE
1	**Cr(VI) safety / PPE / respirator / SDS / hygiene / mist & exhaust controls (all stations)**	____	______	______	_____
2	Station 01 — Inspection & Racking (contact)	____	______	______	_____
3	Station 02 — Clean / Degrease + water-break	____	______	______	_____
4	Station 03 — Etch (light — fatigue/tolerance)	____	______	______	_____
5	Station 04 — Rinse + conductivity	____	______	______	_____
6	Station 05 — Desmut / Deox	____	______	______	_____
7	Station 06 — Rinse	____	______	______	_____
8	Station 07 — Anodize (part = anode; **ramped voltage; Cr VI**)	____	______	______	_____
9	Thickness/cycle control & dimensional growth	____	______	______	_____
10	Station 08 — Post-anodize rinse (open pores; Cr(VI) recovery)	____	______	______	_____
11	Station 09 — Dye (usually N/A)	____	______	______	_____
12	Station 10 — Seal (chromate **(Cr VI)** or hot water)	____	______	______	_____
13	Station 11 — Rinse / Dry	____	______	______	_____
14	Station 12 — Final Inspection (thickness/corrosion/appearance)	____	______	______	_____
15	LOTO on rectifier / electrical safety	____	______	______	_____
16	Emergency response — eyewash, shower, spill, Cr(VI) exposure reporting, fluoride first-aid	____	______	______	_____
17	**Cr(VI) waste treatment (reduce-then-precipitate) & air-permit controls**	____	______	______	_____
18	Troubleshooting — symptom recognition	____	______	______	_____

Operator name: ______________________ **Hire/start date:** ______________

Supervisor: ______________________ **Review cycle:** ☐ Annual ☐ Other: __________

⚠ **HEADS-UP**

Safety rows (1, 15, 16, 17) are not optional and not “fill in later.” An operator may not work *any* station until Cr(VI) safety/PPE/respirator/hygiene, LOTO, emergency response, and Cr(VI) waste-control awareness are signed. **Safety competency precedes production competency** — doubly so on a carcinogen line.

PART FOUR — CERTIFICATION
20 QUESTIONS · PASSING SCORE 80% (16/20)

COMPLETION TEST

COVERS EVERY STATION PLUS SAFETY — ANSWER IN YOUR OWN WORDS WHERE ASKED

★ **REMEMBER**
Passing score: 80% (16 of 20). Any **safety** question answered wrong (**Q2, Q5, Q11, Q14, Q18**) is an automatic fail of the safety gate — re-teach and re-test before sign-off, regardless of total score. The Answer Key follows — no peeking until you're done!

Name: ______________ **Date:** ______________ **Score:** ________ / 20 **Safety-gate Qs: 2, 5, 11, 14, 18**

Q1. Anodizing differs from electroplating mainly because in anodizing the part is the:
A. Cathode, with a different metal deposited on it B. Anode, with oxide grown out of the part itself C. Insulator D. Cathode, and it dissolves

Q2. **[SAFETY GATE]** The Type I chromic anodize bath is:
A. Dilute sulfuric acid B. Sodium hydroxide C. Chromic acid — hexavalent chromium (Cr VI), a confirmed human carcinogen D. Nickel sulfate

Q3. Typical Type I coating thickness is about:
A. 0.5–1 mm B. ~0.5–7.5 µm (often 2–5) C. 25–100 µm D. There is no coating

Q4. Compared with sulfuric Type II, the chromic Type I bath runs:
A. Much colder B. Warmer (~32–40 °C) and on a ramped/stepped voltage C. At constant 200 V D. With no current

Q5. (Short answer) **[SAFETY GATE]** Name the **PEL and Action Level** for Cr(VI) (with the governing OSHA standard) and list **three** controls that keep Cr(VI) mist out of your lungs.

Q6. (Short answer) Explain in your own words why anodizing is **not** plating — name which electrode the part is and **where the coating comes from.**

Q7. The two big reasons aerospace specifies chromic (Type I) over sulfuric on stressed assemblies are:
A. Color and shine B. It doesn't significantly cut fatigue strength, and its trapped residue is far less corrosive in crevices C. It's thicker D. It's harder

Q8. "Dimensional growth" on Type I is:
A. Huge B. Small (because the coating is thin) but not exactly zero — still ~half in/half out C. Negative D. Random

Q9. Type I is usually **not dyed** because:
A. Dye is illegal B. The coating is thin and gray/opaque, so it carries color poorly; it's left natural or used as a paint base C. The pores are too big D. It can't be sealed

Q10. The voltage on a Type I tank is:
A. Slammed straight to full B. Ramped/stepped up gradually (e.g., toward ~22 V or ~40 V per the cycle) C. AC, not DC D. Always 12 V

Q11. (Short answer) **[SAFETY GATE]** Your shop uses a **sodium-dichromate seal.** What additional hazard does that seal carry compared with a hot-water seal, and what does that mean for how you handle it and its waste?

Q12. Two common **chromic-free replacements** being adopted to reduce hex-chrome are:
A. Type II and Type III B. BSAA (Boric-Sulfuric, Type IC) and thin-film sulfuric (TSA) C. Nickel and zinc D. Paint and powder coat

Q13. A part comes out **degraded/streaky** and the lab finds the chromic bath picked up **sulfate.** The likely route in was:
A. Too much DI water B. Drag-in from sulfuric/desmut chemistry through poor rinsing C. The alloy D. The rectifier

Q14. (Short answer) [SAFETY GATE] Describe the **reduce-then-precipitate** method for treating Cr(VI) wastewater (what you do in each step and the goal), and state one place Cr(VI) water must **never** go.

Q15. On Type I, the etch is kept **light** because:
A. To save chemicals only B. Heavy etching removes metal and hurts the tolerance and fatigue life chromic exists to protect C. It must be light to dye
D. It doesn't matter

Q16. (Short answer) Name the **two** common Type I seal methods and give **one tradeoff** of each.

Q17. Coating quality acceptance on a thin gray Type I part usually leads with:
A. Color match B. Corrosion performance (salt spray) and uniform appearance at the spec thickness C. Gloss D. Weight

Q18. (Short answer) [SAFETY GATE] You see **visible mist rising off the chromic anodize tank** and the foam blanket looks gone. What is happening, and what do you do?

Q19. Chloride contamination in the chromic bath causes:
A. Brighter color B. Pitting/burning of the coating C. Faster sealing D. Nothing

Q20. (Short answer) Name **two** ways chromic Type I anodize differs from sulfuric Type II (beyond just the acid).

⚠ **SAFETY SECTION — MUST BE CORRECT TO CERTIFY**

Questions **2, 5, 11, 14, and 18** cover hazards that can permanently harm you or others — including the carcinogenic Cr(VI) bath and seal, mist control, and waste treatment. A miss on any safety item means re-teach and re-test before sign-off.

End of test. Hand to your trainer for grading.

FOR TRAINERS & SELF-CHECKERS

ANSWER KEY

SHORT-ANSWER RESPONSES DON'T NEED TO MATCH WORD-FOR-WORD — LOOK FOR THE KEY CONCEPTS IN BOLD

HEADS-UP

Keep this page out of the trainee's copy. Questions **2, 5, 11, 14, and 18** are safety-critical — any wrong answer there requires re-training on that topic before sign-off, regardless of total score.

Q1. B — The part is the **anode**, and oxide is grown out of the part itself.

Q2. C — **Chromic acid; hexavalent chromium (Cr VI), a confirmed human carcinogen.**

Q3. B — ~0.5–7.5 µm (often 2–5).

Q4. B — Warmer (~32–40 °C) and a **ramped/stepped** voltage.

Q5. PEL = 5 µg/m³; Action Level = 2.5 µg/m³ (8-hr averages, **OSHA 29 CFR 1910.1026**). Three controls (any three): **fume suppressant/foam blanket, local exhaust (push-pull/lateral), fume scrubber, respiratory protection, hygiene/no eating-smoking.** (Engineering controls before PPE.)

Q6. In plating the part is the **cathode** and a **different metal is deposited** from the bath. In anodizing the part is the **anode**, and the coating is **aluminum oxide grown out of the part's own surface** (oxygen from the bath combines with the part's aluminum) — nothing is deposited from elsewhere.

Q7. B — Chromic **doesn't significantly reduce fatigue strength**, and **trapped residue is far less corrosive** in crevices/assemblies than sulfuric.

Q8. B — Small (thin coating) but not exactly zero; still ~half in/half out.

Q9. B — Thin, gray/opaque coating carries color poorly; left natural or used as a **paint base.**

Q10. B — Ramped/stepped up gradually; never slammed to full (which burns the part).

Q11. A dichromate seal **is itself hexavalent chromium (Cr VI)** — so it carries the **same carcinogen/mist/skin hazards** as the anodize bath. Handle with **mist control/ventilation, full PPE/respirator per program**, and route its **waste to Cr(VI) treatment — never to drain.** (A hot-water seal avoids this.)

Q12. B — **BSAA (Boric-Sulfuric, Type IC)** and **thin-film sulfuric (TSA).**

Q13. B — **Sulfate drag-in** from sulfuric/desmut through poor rinsing; sulfate degrades the chromic coating.

Q14. Step 1 — Reduce: add a reducing agent (sodium metabisulfite / SO_2 / ferrous sulfate) at **low pH** to convert **Cr^{6+}** → **Cr^{3+}**. **Step 2 — Precipitate: raise the pH** so trivalent chromium drops out as **chromium-hydroxide sludge**, then settle/filter it; test water before discharge; manage sludge as hazardous waste. Goal: turn the carcinogen into the far-safer trivalent form and remove it. Cr(VI) water must **never reach a floor drain / sewer** (reportable release).

Q15. B — Heavy etching removes metal and **hurts tolerance and fatigue life** — the things chromic exists to protect.

Q16. Dilute dichromate (chromate) seal — best corrosion/inhibitor **but is Cr(VI)**; **hot DI water seal** — chromium-free/simple **but high energy and bloom-prone if water/pH off.**

Q17. B — **Corrosion performance (salt spray) and uniform appearance** at the spec thickness (not color).

Q18. The **mist suppression has failed**, so you are being **over-exposed to carcinogenic Cr(VI) mist. Stop running the tank, leave the immediate area / don respirator per program, report it** so suppression/exhaust is restored and a Cr(VI) exposure check is done. Do **not** keep running.

Q19. B — Pitting/burning. Keep chloride very low (use DI water).

Q20. Any two: chromic is **thin** (vs. medium); **warm bath** (vs. cold); **ramped voltage** (vs. constant); **gray/rarely dyed** (vs. brightly dyeable); **low fatigue debit / crevice-safe** (vs. higher); **Cr(VI) carcinogen bath** (vs. sulfuric); **steel/lead cathodes**; **chromate or hot-water seal.**

★ **REMEMBER**

16/20 passes — but **every safety question (2, 5, 11, 14, 18) must be correct to certify.** Miss a safety item → re-teach and re-test before sign-off. We don’t gamble with people, and on a Cr(VI) line that rule is absolute.

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CERTIFICATE OF COMPLETION

CHROMIC ACID ANODIZING (TYPE I) TRAINING PROGRAM

This certifies that

____

OPERATOR NAME

has successfully completed the **Chromic Acid Anodizing (Type I)** training course — covering the full anodize workflow (Inspection & Racking, Clean, Light Etch, Rinse, Desmut/Deox, Chromic Anodize, Rinse, Dye [usually N/A], Seal, Rinse/Dry, and Final Inspection), the fundamental concept that **the part is the anode and the oxide is grown out of the part itself**, the thin gray paint-base oxide and its ramped-voltage chromic chemistry, minimal dimensional growth, the fatigue-strength and crevice/assembly advantages that make Type I the aerospace anodize, the Type I/IB/IC family and the move toward BSAA/TSA, aluminum-alloy effects, chromate vs. hot-water sealing, and — **above all — the safety practices required on a hexavalent-chromium line:** Cr(VI) is a confirmed human carcinogen; mist suppression, ventilation, scrubber, respiratory protection, hygiene, OSHA 29 CFR 1910.1026 (PEL 5 μg/m³ / action level 2.5 μg/m³), nasal-septum and chrome-ulcer hazards, emergency response, and **reduce-then-precipitate Cr(VI) waste treatment** — and has passed the Completion Test with a score of ________ / 20, including all required safety competencies.

This operator is hereby cleared for **independent work** at the certified stations listed on the Operator Training Matrix.

Date of completion: ____________________ **Re-certification due:** ____________________

______________________________ ______________________________

TRAINEE SIGNATURE CERTIFYING TRAINER

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SUPERVISOR

Training reference only. Verify all parameters against your chemistry supplier's TDS, customer/aerospace specifications (e.g., MIL-PRF-8625 Type I / IB), and applicable regulatory requirements before production use. **Hexavalent chromium (Cr VI) chromic acid is a confirmed human carcinogen** — always consult current SDS and comply with OSHA 29 CFR 1910.1026, EPA, REACH, and local regulations.